

Reproductive Systems and Lower Limb Anatomy

Courtesy of Dr. M. Evans

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Contents

I	The Human Pelvis	6
1	The Bony Pelvis	6
1.1	The Innominate Bone (Hip Bone)	6
1.2	The Pelvic Spine	7
1.3	Joints and Ligaments of the Pelvis	7
2	Sexual Dimorphism of the Pelvis: An Obstetric and Evolutionary Perspective	9
2.1	General and Specific Structural Differences	9
2.2	The Caldwell-Moloy Classification	10
3	The Pelvic Walls and Floor: Musculature and Fascial Support Systems	11
3.1	Muscles of the Pelvic Walls	11
3.2	The Pelvic Diaphragm (Pelvic Floor)	11
3.3	Fascial and Connective Tissue Support	12
4	Neurovasculature of the Pelvis: Pathways and Territories	13
4.1	Arterial Supply	14
4.2	Venous Drainage	15
4.3	Nerve Supply	15
5	Applied Anatomy and Clinical Correlates of the Pelvis	17
5.1	General Clinical Correlates	17
5.2	Special Focus: Obstetric Pelvimetry and the Mechanism of Labor	17
II	The Female Reproductive System: Anatomy, Histology, and Embryology	19

6	Gross and Applied Anatomy of the Internal and External Genitalia	20
6.1	Internal Genitalia	20
6.2	External Genitalia (Vulva)	21
7	Microscopic Anatomy (Histology) of the Female Reproductive Organs	22
8	Clinical Correlates of the Female Reproductive System	23
III	The Male Reproductive System: Anatomy, Histology, and Embryology	24
9	Gross and Applied Anatomy of the Internal and External Genitalia	24
9.1	External Genitalia	24
9.2	Internal Genitalia	24
10	Microscopic Anatomy (Histology) of the Male Reproductive Organs	26
11	Clinical Correlates of the Male Reproductive System	27
IV	Embryological Development, Homologies, and Congenital Anomalies	28
12	Development of the Reproductive Systems: From Indifferent to Differentiated	28
13	Homologous Structures of the Male and Female Reproductive Systems	29
14	Common Congenital Anomalies and Their Embryological Basis	30
14.1	Female Congenital Anomalies	30
14.2	Male Congenital Anomalies	30
14.3	Disorders of Sex Development (DSD)	31
V	The Bony Framework and Articulations	31

15 Osteology: The Foundation of the Lower Limb	32
15.1 The Pelvic Girdle: The Connection to the Axial Skeleton	32
15.2 The Femur and Patella: The Thigh and Knee Cap	33
15.3 The Tibia and Fibula: The Bones of the Leg	34
15.4 The Bones of the Foot: Tarsals, Metatarsals, and Phalanges	35
16 Arthrology: The Joints of the Lower Limb	35
16.1 The Hip Joint (Coxal Joint)	36
16.2 The Knee Joint	36
16.3 The Ankle and Subtalar Joints: Adaptation and Support	37
16.4 The Arches of the Foot: Engineering for Weight Distribution	38
 VI The Muscular System and Functional Compartments	 39
17 Myology of the Hip and Thigh	39
17.1 The Gluteal Region: Extensors, Abductors, and Rotators	39
17.2 The Anterior Thigh: The Quadriceps Femoris Group	41
17.3 The Medial Thigh: The Adductor Group	42
17.4 The Posterior Thigh: The Hamstring Group	44
18 Myology of the Leg and Foot	45
18.1 The Anterior and Lateral Compartments of the Leg	45
18.2 The Posterior Compartment of the Leg	46
18.3 The Intrinsic Muscles of the Foot	48
19 Key Anatomical Spaces and Clinical Implications	49
19.1 The Femoral Triangle, Adductor Canal, and Popliteal Fossa	50
19.2 Compartment Syndrome: Pathophysiology and Anatomical Basis	50
19.3 Trendelenburg's Sign: The Anatomy of Pelvic Stabilization	50

VII	The Neurovascular Network	51
20	Innervation of the Lower Limb	51
20.1	The Lumbosacral Plexus: Formation and Major Nerves	51
20.2	Course and Distribution of Major Nerves	52
20.3	Cutaneous Innervation and Dermatome Mapping	53
21	Vasculature of the Lower Limb	54
21.1	Arterial Supply: From the Femoral Artery to the Plantar Arch	54
21.2	Venous Drainage: The Superficial and Deep Systems	54
21.3	Lymphatics	55
VIII	Integrated Clinical Anatomy and Procedures	55
22	Common Pathologies: An Anatomical Perspective	55
22.1	Fractures and Dislocations	55
22.2	Soft Tissue Injuries	56
22.3	Systemic and Overuse Conditions	56
23	Applied Surface Anatomy and Clinical Procedures	57
23.1	Surface Anatomy and Palpation	57
23.2	Clinical Examination Tests	57
23.3	Pulse Points	58
23.4	Intramuscular Injections	58

Part I

The Human Pelvis

The human pelvis represents a complex and highly specialized region of the body, serving as the critical junction between the axial skeleton and the lower limbs. It is responsible for supporting the weight of the upper body, facilitating locomotion, and providing protection for the pelvic and abdominopelvic viscera, including the urinary bladder, rectum, and the internal reproductive organs. Its structure is a remarkable example of functional adaptation, particularly evident in the profound sexual dimorphism that accommodates the biomechanical demands of childbirth in females. This section provides a detailed examination of the bony pelvis, its sexual differences, the supporting musculature and fascia, and its intricate neurovascular supply, culminating in an analysis of its applied anatomy and clinical significance.

1 The Bony Pelvis

The bony pelvis, or pelvic skeleton, is a robust ring-like structure formed by the articulation of four bones: the two hip bones (ossa coxae or innominate bones) laterally and anteriorly, and the sacrum and coccyx posteriorly. The hip bone itself is a composite structure, formed by the post-pubertal fusion of three distinct bones—the ilium, ischium, and pubis—at a Y-shaped cartilaginous junction within the acetabulum known as the triradiate cartilage. This developmental origin is clinically significant, as pediatric pelvic trauma may result in injury at these cartilaginous growth plates rather than through ossified bone. The complete structure is often divided into the pelvic girdle (the two hip bones) and the pelvic spine (sacrum and coccyx), which together form a basin that connects the trunk to the lower extremities.

1.1 The Innominate Bone (Hip Bone)

Each innominate bone is a large, irregularly shaped bone with distinct parts and numerous landmarks crucial for muscle attachment and clinical reference. **Ilium:** The ilium is the largest and most superior of the three components, forming the "wings" of the pelvis. Its superior margin is the palpable **iliac crest**, which arches from the **anterior superior iliac spine (ASIS)** in the front to the **posterior superior iliac spine (PSIS)** in the back. The ASIS is a vital surface landmark for locating the inguinal ligament, the femoral nerve, and for performing various nerve blocks. The PSIS is often visible as a "dimple" on the lower back and serves as an attachment point for strong ligaments supporting the sacroiliac joint. Inferior to these are the **anterior inferior iliac spine (AIIS)** and **posterior**

inferior iliac spine (PIIS). The internal, concave surface of the iliac wing is the **iliac fossa**, which provides a large origin for the iliacus muscle. The posteromedial surface features the ear-shaped **auricular surface**, which articulates with the sacrum, and the roughened **iliac tuberosity** for ligamentous attachment. Running obliquely forward from the auricular surface is the **arcuate line**, a smooth ridge that forms part of the pelvic brim—the boundary between the greater (false) and lesser (true) pelvis. **Ischium:** The ischium forms the strong posteroinferior part of the hip bone and bears the body's weight during sitting. Its most prominent feature is the large, roughened **ischial tuberosity**, which serves as the attachment site for the hamstring muscles. Projecting postero-medially is the sharp **ischial spine**, a critical obstetric landmark for assessing fetal descent and the target for pudendal nerve anesthesia. The ischial spine separates the **greater sciatic notch** above from the **lesser sciatic notch** below. The **ischial ramus** projects anteriorly to fuse with the inferior ramus of the pubis. **Pubis:** The pubis is the anteroinferior part of the hip bone. It consists of a body and superior and inferior rami. The bodies of the left and right pubic bones articulate in the midline at the **pubic symphysis**. The superior ramus contributes to the acetabulum and features the **pectineal line**, which continues from the arcuate line of the ilium to complete the pelvic brim. The conjoined inferior pubic ramus and ischial ramus form the **ischiopubic ramus**, which defines the anterolateral boundary of the pelvic outlet. The pubis and ischium together encircle the large **obturator foramen**, which is mostly closed by the obturator membrane, providing a surface for muscle attachment while allowing passage for the obturator nerve and vessels. **Acetabulum:** This deep, cup-shaped socket on the lateral aspect of the hip bone is where the ilium, ischium, and pubis fuse. It articulates with the head of the femur to form the hip joint. Its articular surface is C-shaped and lined with hyaline cartilage, and its depth is enhanced by a fibrocartilaginous rim, the **acetabular labrum**.

1.2 The Pelvic Spine

Sacrum: A triangular bone formed by the fusion of five sacral vertebrae. Its superior surface, or base, articulates with the fifth lumbar vertebra at the lumbosacral joint, and its anterior projecting edge is the **sacral promontory**, a key landmark of the pelvic inlet. Laterally, its wing-like alae articulate with the iliac bones at the sacroiliac joints. **Coccyx:** The small tailbone, typically formed from four fused vertebrae, articulates with the inferior apex of the sacrum. It provides an attachment point for parts of the pelvic floor musculature.

1.3 Joints and Ligaments of the Pelvis

The bones of the pelvis are bound together by some of the strongest joints and ligaments in the body, ensuring stability for weight transmission. **Sacroiliac (SI) Joints:** These are strong, weight-bearing compound joints, having both synovial and fibrous components, that link the axial skeleton to the pelvic girdle. They permit only slight gliding and rotational movements. They are stabilized by the powerful **anterior, posterior, and interosseous sacroiliac ligaments**. The interosseous ligament is the strongest of these, binding

the sacral and iliac tuberosities together. **Pubic Symphysis:** This is a secondary cartilaginous joint (symphysis) where the bodies of the pubic bones are joined by a fibrocartilaginous disc. It is essentially immobile, except during pregnancy when hormonal changes induce laxity, allowing for a slight increase in pelvic diameter during childbirth. **Accessory Ligaments:** Two major ligaments, the **sacrospinous ligament** (from the sacrum to the ischial spine) and the **sacrospinous ligament** (from the sacrum to the ischial spine), are crucial for stabilizing the sacrum and preventing its upward tilting. Functionally, they convert the greater and lesser sciatic notches into foramina, creating vital passageways for structures like the sciatic nerve and pudendal neurovascular bundle to exit and enter the pelvis. The architectural arrangement of the bony pelvis and its ligamentous connections forms a closed kinetic ring. This structural principle is fundamental to its biomechanical function and its response to trauma. Because it is a closed loop, a traumatic force sufficient to cause a fracture or dislocation at one point in the ring will almost certainly be transmitted through the structure, resulting in a second point of disruption elsewhere. A single break is mechanically unstable and highly improbable. This concept fundamentally alters the clinical evaluation of pelvic injuries. For instance, the identification of an anterior injury, such as a pubic ramus fracture or a diastasis of the pubic symphysis (an "open book" fracture), mandates a thorough search for a corresponding posterior injury, like a sacral fracture or sacroiliac joint disruption. The posterior injury is often the primary determinant of pelvic stability and the source of major, life-threatening hemorrhage from the posterior venous plexuses. Therefore, the clinical assessment must move beyond identifying a single fracture to evaluating the integrity of the entire ring, which dictates whether management will be conservative or require urgent surgical stabilization.

Landmark	Bone of Origin	Description and Anatomical Relations	Clinical/Functional Significance
Iliac Crest	Ilium	The superior, arching border of the iliac wing.	Palpable at the waistline; site of bone marrow biopsy; attachment for abdominal and gluteal muscles.
Anterior Superior Iliac Spine (ASIS)	Ilium	The rounded anterior termination of the iliac crest.	Key palpable landmark; attachment for inguinal ligament and sartorius muscle.
Posterior Superior Iliac Spine (PSIS)	Ilium	The posterior termination of the iliac crest.	Often marked by a skin dimple; attachment for sacroiliac ligaments.
Greater Sciatic Notch	Ilium/Ischium	A large indentation on the posterior margin, superior to the ischial spine.	Converted to greater sciatic foramen by sacrospinous ligament; passage for piriformis muscle, sciatic nerve, gluteal vessels/nerves, and pudendal vessels/nerve.

Landmark	Bone of Origin	Description and Anatomical Relations	Clinical/Functional Significance
Ischial Spine	Ischium	A sharp, bony projection separating the greater and lesser sciatic notches.	Attachment for sacrospinous ligament; key obstetric landmark for assessing fetal station ("0 station"); landmark for pudendal nerve block.
Ischial Tuberosity	Ischium	A large, roughened prominence on the posteroinferior ischium.	Supports body weight when sitting ("sit bone"); attachment for hamstring muscles and sacrotuberous ligament.
Pubic Symphysis	Pubis	The midline cartilaginous joint between the two pubic bodies.	Articulates the two hip bones anteriorly; relaxes during pregnancy to increase pelvic diameter.
Sacral Promontory	Sacrum	The anteriorly projecting superior margin of the first sacral vertebra.	Key obstetric landmark defining the posterior boundary of the pelvic inlet for the anteroposterior (conjugate) diameter.

2 Sexual Dimorphism of the Pelvis: An Obstetric and Evolutionary Perspective

The male and female pelves exhibit significant structural differences, a phenomenon known as sexual dimorphism. While some of this variation is related to overall body size, the most conspicuous differences are shape-related adaptations in the female pelvis to facilitate parturition. These distinctions, which become most pronounced during the hormonally driven growth spurt of adolescence, reflect a delicate evolutionary compromise.

2.1 General and Specific Structural Differences

In general, the female pelvis is wider, shallower, and lighter in structure compared to the male pelvis, which is taller, narrower, and more robust to support a heavier build and larger muscles. **Greater (False) Pelvis:** The female false pelvis is shallower, while the male's is deeper, a result of the iliac crests being higher and more pronounced in males. **Lesser (True) Pelvis:** The shape of the true pelvis is the most critical distinction. **Pelvic Inlet:** The female inlet is typically wider and oval or round in shape, providing a spacious entry for the fetal head. In contrast, the male inlet is narrower and heart-shaped (android) due to the projection of the sacral promontory into the

aperture. **Pelvic Cavity:** The female pelvic cavity is wide and shallow with parallel side walls. The male cavity is deeper, narrower, and funnel-shaped, with walls that converge towards the outlet. **Pelvic Outlet:** The female outlet is larger. The ischial tuberosities are farther apart, and the ischial spines are less prominent, creating a wider transverse diameter. In males, these structures project further medially, constricting the outlet. **Component Bones: Sacrum:** The female sacrum is shorter, wider, and more curved posteriorly, increasing the posterior space in the true pelvis. The male sacrum is longer, narrower, and straighter. **Coccyx:** The female coccyx is more mobile and angled less anteriorly, allowing it to be pushed back during childbirth. The male coccyx is less mobile and angled more anteriorly. **Pubic Arch and Subpubic Angle:** The angle formed by the inferior ischiopubic rami is wide and obtuse (greater than 90 degrees) in females, forming a rounded pubic arch. In males, this angle is narrow and acute (less than 90 degrees), forming a V-shaped subpubic angle. **Other Features:** The female acetabula are farther apart and face more anteriorly, which contributes to a wider stance and a characteristic gait with more hip rotation. The female obturator foramen is more triangular or oval, whereas it is rounder in males.

2.2 The Caldwell-Moloy Classification

This system categorizes the female pelvis into four ideal shapes based on the morphology of the pelvic inlet, which has implications for the mechanism of labor. **Gynecoid (approx. 50% of women):** The classic female pelvis. The inlet is round or slightly oval, with a generous transverse diameter. The side walls are straight, the ischial spines are not prominent, and the pubic arch is wide. This shape is the most favorable for a straightforward vaginal delivery. **Android (approx. 20%):** The male-type pelvis. The inlet is heart-shaped or triangular, with the posterior segment narrowed by a prominent sacrum. The side walls converge, the ischial spines are prominent, and the pubic arch is narrow. This shape is associated with a higher risk of difficult labor (dystocia) and may necessitate an operative delivery. **Anthropoid (approx. 25%):** An ape-like pelvis. The inlet is a long anteroposterior oval. This shape favors engagement of the fetal head in an occiput-posterior position, which can lead to a longer labor. **Platypelloid (approx. 5%):** A flat pelvis. The inlet is a wide transverse oval with a very short anteroposterior diameter. The fetal head may have difficulty engaging in the inlet, often leading to the need for a Cesarean section.

It is important to recognize that most pelvises are of a mixed type rather than fitting perfectly into one of these categories. The profound sexual dimorphism of the pelvis is the result of a significant evolutionary conflict, often termed the "obstetrical dilemma". On one hand, the mechanics of efficient bipedal locomotion favor a narrow pelvis to position the hip joints directly under the body's center of gravity, minimizing energy expenditure during walking. On the other hand, the evolution of humans towards greater encephalization has resulted in a fetus with a disproportionately large head, which requires a wide and spacious pelvic canal for successful parturition. The female pelvis, particularly the gynecoid type, represents the optimal evolutionary compromise between these two opposing selective pressures. This inherent tension explains why human childbirth is a complex biomechanical process. The fetal head cannot pass through the pelvic canal in a straight line; it must perform a series of intricate rotations—the cardinal movements of labor—to align its smallest

diameters with the largest diameters of each successive pelvic plane. This evolutionary context provides the ultimate rationale for the clinical importance of pelvimetry and a deep understanding of the mechanism of labor.

3 The Pelvic Walls and Floor: Musculature and Fascial Support Systems

While the bony pelvis provides the rigid framework, the pelvic cavity is enclosed and supported by a complex arrangement of muscles and fascia that form its walls and floor. These soft tissue structures are not merely passive liners but constitute a dynamic system essential for supporting the pelvic viscera, maintaining urinary and fecal continence, and facilitating functions such as childbirth and sexual activity.

3.1 Muscles of the Pelvic Walls

The walls of the true pelvis are lined by two main muscles, which primarily function as lateral rotators of the thigh at the hip joint but whose bodies form the muscular bed for the pelvic organs and neurovasculature. **Piriformis:** This flat, pyramidal muscle originates from the anterior surface of the sacrum and sacrotuberous ligament. Its fibers converge to pass through the greater sciatic foramen, forming the muscular posterolateral wall of the pelvic cavity before inserting on the greater trochanter of the femur. The sacral plexus lies on its anterior surface. **Obturator Internus:** This fan-shaped muscle originates from the margins of the obturator foramen and the internal surface of the obturator membrane. It forms the anterolateral wall of the pelvis. Its tendon exits the pelvis through the lesser sciatic foramen to insert on the greater trochanter. The levator ani muscles take part of their origin from the fascia covering the obturator internus.

3.2 The Pelvic Diaphragm (Pelvic Floor)

The pelvic floor, or pelvic diaphragm, is a broad, funnel-shaped muscular sling that forms the inferior boundary of the pelvic cavity, separating it from the perineum below. It is pierced by the urogenital hiatus anteriorly (for the urethra and vagina) and the anal hiatus posteriorly (for the anal canal). It is composed of the levator ani and coccygeus muscles. **Levator Ani:** This is the largest and most functionally significant component of the pelvic floor. It is a broad sheet of muscle composed of three paired parts that work in unison. **Puborectalis:** Arising from the posterior surface of the pubic bodies, its fibers form a U-shaped sling that loops around the posterior aspect of the anorectal junction. The tonic contraction of this muscle pulls the junction anteriorly, creating the anorectal angle (approximately 80 degrees), which is the primary mechanism for maintaining fecal continence. During defecation, the puborectalis voluntarily relaxes, allowing the anorectal angle to straighten. **Pubococcygeus:** Forming the main, intermediate part of the levator ani, this muscle arises from the body of the pubis and the anterior part of the tendinous arch of the obturator fascia. Its

fibers travel posteriorly to insert into the anococcygeal body and the coccyx. Medial fibers of the pubococcygeus, collectively known as the **pubovisceralis**, blend with the walls of the urethra (pubourethralis), vagina (pubovaginalis), and anal canal (puboanalis), providing active support and sphincteric function to these structures. **Iliococcygeus:** This is the thin, posterolateral portion of the levator ani. It arises from the posterior part of the tendinous arch and the ischial spine, and its fibers blend in the midline to form a raphe that inserts into the anococcygeal body and coccyx. It functions to elevate and support the pelvic floor. **Coccygeus (Ischiococcygeus):** A small, triangular muscle situated posterior to the levator ani, lying on the pelvic surface of the sacrospinous ligament. It extends from the ischial spine to the lateral margins of the lower sacrum and coccyx. It assists the levator ani in supporting the pelvic viscera and flexes the coccyx.

Innervation for the pelvic diaphragm is supplied primarily by direct branches from the anterior rami of S3 and S4 (the nerve to levator ani) and by perineal branches of the pudendal nerve (S2-S4).

3.3 Fascial and Connective Tissue Support

Endopelvic Fascia: This is a continuous sheet of connective tissue that lines the pelvic cavity, covering the muscles of the pelvic walls and floor (parietal fascia) and investing the pelvic organs (visceral fascia). Condensations of this fascia form critical supportive ligaments, such as the cardinal and uterosacral ligaments in the female, which anchor the cervix and uterus to the pelvic walls. **Perineal Body:** Also known as the central tendon of the perineum, this is a fibromuscular node in the midline between the vaginal orifice and the anal canal. It is a convergence point for the bulbospongiosus, superficial transverse perineal muscles, and fibers from the external anal sphincter and levator ani. Its integrity is crucial for the support of the posterior vaginal wall and the entire pelvic floor, and it is often damaged during childbirth. The pelvic floor should not be viewed as a simple, static hammock. It is a highly dynamic structure that functions as the base of the "core" musculature, working in an integrated system with the transverse abdominis, multifidus, and respiratory diaphragm to regulate intra-abdominal pressure. For example, during a cough or sneeze, a sudden increase in intra-abdominal pressure is generated by the contraction of the diaphragm and abdominal muscles. A healthy pelvic floor reflexively contracts in anticipation of this pressure wave, elevating and tensing to maintain closure of the urethral and anal sphincters, thereby ensuring continence. Pelvic floor dysfunction, such as stress urinary incontinence or pelvic organ prolapse, is therefore not merely a result of weak pelvic muscles but represents a failure of this coordinated neuromuscular system. This understanding has shifted the clinical management of such conditions from a focus solely on isolated pelvic floor exercises (Kegel exercises) to a more holistic approach involving comprehensive core stabilization, breathing mechanics, and postural training.

Muscle	Origin	Insertion	Innervation	Primary Action(s)
Levator Ani: Puborectalis	Posterior aspect of the bodies of the pubic bones	Forms a U-shaped sling posterior to the anorectal junction, blending with the contralateral muscle	Nerve to levator ani (S4); Pudendal nerve (S2-S4)	Maintains anorectal angle for fecal continence; relaxes during defecation
Levator Ani: Pubococcygeus	Posterior aspect of the pubic bones, anterior tendinous arch	Anococcygeal body, coccyx, perineal body, and walls of pelvic viscera	Nerve to levator ani (S3-S4); Pudendal nerve branches	Elevates and supports pelvic floor and viscera; assists in sphincteric control
Levator Ani: Iliococcygeus	Tendinous arch of obturator internus fascia, ischial spine	Anococcygeal body and coccyx	Nerve to levator ani (S3-S4); Pudendal nerve branches	Elevates and supports pelvic floor and anorectal canal
Coccygeus (Ischiococcygeus)	Ischial spine	Inferior end of sacrum and coccyx	Anterior rami of S4 and S5	Supports pelvic viscera; flexes coccyx
Piriformis	Anterior surface of sacrum (S2-S4 segments)	Superior border of greater trochanter of femur	Branches from anterior rami of S1 and S2	Laterally rotates and abducts thigh; forms posterolateral pelvic wall
Obturator Internus	Internal surface of obturator membrane and surrounding bone	Medial surface of greater trochanter of femur	Nerve to obturator internus (L5, S1, S2)	Laterally rotates and abducts thigh; forms anterolateral pelvic wall

4 Neurovasculature of the Pelvis: Pathways and Territories

The pelvic cavity houses a dense and complex network of arteries, veins, lymphatics, and nerves that supply the pelvic walls, floor, viscera, and perineum. A thorough understanding of their courses, relationships, and common variations is essential for surgical procedures and for interpreting pelvic pathology.

4.1 Arterial Supply

The majority of the arterial supply to the pelvis originates from the **internal iliac artery**, also known as the hypogastric artery. **Origin and Course:** The abdominal aorta bifurcates at the L4 vertebral level into the right and left common iliac arteries. Each common iliac artery then divides, typically at the level of the sacroiliac joint, into the external and internal iliac arteries. The external iliac artery continues along the pelvic brim to supply the lower limb, while the internal iliac artery descends into the lesser pelvis to supply its contents. At the superior margin of the greater sciatic foramen, it typically divides into an anterior and a posterior division or trunk. **Posterior Division:** This division primarily gives rise to parietal branches that supply the pelvic wall and gluteal region. **Iliolumbar Artery:** Ascends in a recurrent fashion to supply the iliacus muscle, psoas major, quadratus lumborum, and the iliac fossa. **Lateral Sacral Arteries:** Usually present as superior and inferior branches, they descend anterior to the sacrum to supply the sacral canal, erector spinae muscles, and overlying skin. **Superior Gluteal Artery:** The largest branch of the posterior division, it exits the pelvis through the greater sciatic foramen superior to the piriformis muscle to become the major blood supply to the gluteal muscles. **Anterior Division:** This division provides both visceral and parietal branches, supplying the pelvic organs, perineum, and parts of the gluteal and thigh regions. Its branching pattern is notoriously variable. **Umbilical Artery:** In the fetus, this is a large artery carrying deoxygenated blood to the placenta. Postnatally, its proximal part remains patent and gives rise to the **superior vesical arteries**, which supply the superior aspect of the bladder. Its distal part obliterates to form the medial umbilical ligament. **Obturator Artery:** Courses along the lateral pelvic wall with the obturator nerve to exit through the obturator canal, supplying the adductor muscles of the thigh. In a significant percentage of individuals (around 20%), it has an anomalous origin from the inferior epigastric artery (a branch of the external iliac), creating an "aberrant obturator artery" or "corona mortis" (crown of death), which is vulnerable to injury during hernia repair. **Inferior Vesical Artery (in males):** Supplies the base of the bladder, prostate, and seminal vesicles. **Vaginal Artery (in females):** The homologue of the inferior vesical artery, supplying the vagina and adjacent parts of the bladder and rectum. **Uterine Artery (in females):** A large, tortuous artery that runs medially in the base of the broad ligament. It famously crosses superior to the ureter near the cervix before ascending along the lateral border of the uterus. **Middle Rectal Artery:** Arises to supply the middle and lower parts of the rectum, anastomosing with the superior and inferior rectal arteries. **Internal Pudendal Artery:** This is the primary artery of the perineum. It exits the pelvis through the greater sciatic foramen, curves around the ischial spine and sacrospinous ligament, and re-enters the perineum through the lesser sciatic foramen, accompanied by the pudendal nerve. **Inferior Gluteal Artery:** A large terminal branch that exits the pelvis through the greater sciatic foramen inferior to the piriformis muscle to supply the gluteal muscles and hip joint. Other arteries contributing to the pelvis include the **ovarian/testicular artery** (from the abdominal aorta), the **median sacral artery** (from the aortic bifurcation), and the **superior rectal artery** (the terminal branch of the inferior mesenteric artery).

4.2 Venous Drainage

Venous drainage from the pelvis largely mirrors the arterial supply. Tributaries corresponding to the branches of the internal iliac artery drain into the **internal iliac vein**. The pelvis is characterized by extensive, interconnected, and largely valveless venous plexuses (e.g., vesical, prostatic, uterine, vaginal, rectal) that surround the organs. These plexuses ultimately drain into the internal iliac vein. The internal iliac vein joins the **external iliac vein** (which drains the lower limb) to form the **common iliac vein** at the level of the SI joint. The right and left common iliac veins unite at approximately L5 to form the inferior vena cava. The valveless nature of these plexuses and their connections to the vertebral venous plexus (Batson's plexus) provide a pathway for the metastasis of pelvic cancers (e.g., prostate cancer) to the spine and brain.

4.3 Nerve Supply

The innervation of the pelvis is a complex mix of somatic (motor and sensory) and autonomic (sympathetic and parasympathetic) nerves.

Somatic Plexuses: Sacral Plexus: Formed by the lumbosacral trunk (anterior rami of L4-L5) and the anterior rami of S1-S4, it lies on the anterior surface of the piriformis muscle. It provides most of the innervation to the lower limb (via the sciatic nerve), the gluteal region (via the superior and inferior gluteal nerves), and the perineum (via the pudendal nerve).

Pudendal Nerve (S2-S4): As the chief nerve of the perineum, it follows the internal pudendal artery on its course out of the greater sciatic foramen and into the lesser sciatic foramen to enter the pudendal canal. It provides motor supply to the external anal sphincter, external urethral sphincter, and perineal muscles, and sensory supply to the skin of the perineum and external genitalia.

Coccygeal Plexus: A small plexus from S4, S5, and the coccygeal nerve that supplies the skin over the coccyx.

Autonomic Nerves: These nerves control visceral function.

Sympathetic Supply: Originates from the T12-L2 spinal segments. Fibers travel via the **superior hypogastric plexus** (located over the aortic bifurcation) and the paired **hypogastric nerves** to enter the pelvis. They are joined by fibers from the sacral sympathetic trunks (sacral splanchnic nerves). Sympathetic stimulation causes vasoconstriction, contraction of the internal urethral sphincter, and emission (movement of semen into the urethra).

Parasympathetic Supply: Originates from S2-S4 as the **pelvic splanchnic nerves** (nervi erigentes). These fibers are responsible for vasodilation (leading to erection), bladder contraction for micturition, and rectal contraction for defecation.

Inferior Hypogastric (Pelvic) Plexus: This is a large, fan-shaped network of nerves situated on the lateral wall of the pelvis, next to the rectum. It is the crucial distribution center where sympathetic and parasympathetic fibers mix before being distributed to all the pelvic organs via subsidiary plexuses (e.g., prostatic, vesical, uterovaginal). A critical anatomical relationship with profound surgical implications is the crossing of the ureter by the uterine artery in females and the ductus deferens in males, often remembered by the mnemonic "water under the bridge". In females, as the uterine artery courses medially toward the uterus, it passes directly superior to the ureter near the lateral fornix of the vagina. During a hysterectomy, the uterine artery must be clamped and ligated. Due to this intimate proximity, the ureter is at significant risk of being inadvertently clamped, ligated, or transected, which can lead to severe

complications such as a ureterovaginal fistula or hydronephrosis. Similarly, in males, the ductus deferens loops over the ureter as it descends to the back of the bladder. A surgeon must meticulously identify and preserve the ureter during any pelvic procedure involving these structures to prevent iatrogenic injury. This anatomical relationship is a prime example of how a detailed understanding of gross anatomy is directly linked to surgical technique and patient safety.

Division	Artery Name	Primary Structures Supplied	Key Clinical Notes
Posterior	Iliolumbar Artery	Iliacus, psoas major, quadratus lumborum muscles, iliac fossa	Ascends retroperitoneally
Posterior	Lateral Sacral Arteries	Sacral canal structures, erector spinae, skin over sacrum	Run medial to anterior sacral foramina
Posterior	Superior Gluteal Artery	Gluteus medius, gluteus minimus, and tensor fasciae latae muscles	Largest branch; exits pelvis via greater sciatic foramen superior to piriformis
Anterior	Umbilical Artery	Superior aspect of the urinary bladder (via superior vesical arteries)	Distal part obliterates to form the medial umbilical ligament
Anterior	Obturator Artery	Adductor muscles of the thigh, pelvic muscles	Exits via obturator canal; aberrant origin from inferior epigastric a. ("corona mortis") in ~20% of cases
Anterior	Inferior Vesical Artery (Male)	Bladder base, prostate, seminal vesicles, ductus deferens	Homologous to the vaginal artery in females
Anterior	Uterine Artery (Female)	Uterus, cervix, upper vagina, fallopian tube, ovary	Crosses superior to the ureter ("water under the bridge"); highly tortuous
Anterior	Vaginal Artery (Female)	Vagina, bladder base, rectum	Often arises from the uterine artery instead of directly from the internal iliac
Anterior	Middle Rectal Artery	Middle and lower rectum	Anastomoses with superior and inferior rectal arteries
Anterior	Internal Pudendal Artery	Perineum, external genitalia, anal canal	Main artery of the perineum; exits via greater sciatic foramen, re-enters via lesser sciatic foramen
Anterior	Inferior Gluteal Artery	Gluteus maximus muscle, hip joint	Exits pelvis via greater sciatic foramen inferior to piriformis

5 Applied Anatomy and Clinical Correlates of the Pelvis

The integrated anatomy of the pelvis has direct relevance to numerous clinical conditions and procedures. Its role as a bony canal for childbirth, in particular, makes a detailed understanding of its dimensions and biomechanics paramount in obstetrics.

5.1 General Clinical Correlates

Pelvic Fractures: As previously discussed, the stability of the pelvic ring is the key determinant in management. Stable fractures (e.g., isolated pubic ramus fracture) can often be managed conservatively, whereas unstable fractures (e.g., "open book" or vertical shear injuries) are surgical emergencies requiring fixation to restore the ring's integrity and control hemorrhage. **Pelvic Organ Prolapse (POP):** Damage to the levator ani muscles or the endopelvic fascia, most commonly during vaginal childbirth, can compromise the support system for the pelvic organs. This can lead to the descent of the bladder (cystocele), urethra (urethrocele), rectum (rectocele), or uterus/vaginal vault into the vaginal canal. **Pudendal Nerve Block and Entrapment:** The pudendal nerve's course around the ischial spine makes this a reliable landmark for administering a local anesthetic block to achieve perineal analgesia for childbirth or minor surgical procedures. This same anatomical confinement between the sacrospinous and sacrotuberous ligaments can also lead to nerve entrapment, causing chronic perineal pain (pudendal neuralgia), which is characteristically worsened by sitting. **Spread of Pelvic Infection:** The pelvic cavity is continuous with the abdominal cavity. Pelvic inflammatory disease (PID), an infection of the female upper reproductive tract, can lead to the spread of pathogens into the peritoneal cavity. This spread often follows the paracolic gutters, particularly on the right side, leading to perihepatic inflammation (Fitz-Hugh-Curtis syndrome). Furthermore, fluid or pus can collect in the lowest point of the female peritoneal cavity, the **recto-uterine pouch (of Douglas)**, which is accessible for drainage (culdocentesis) via the posterior vaginal fornix.

5.2 Special Focus: Obstetric Pelvimetry and the Mechanism of Labor

The process of childbirth (parturition) is a mechanical feat involving the passage of the fetus (the "passenger") through the maternal bony pelvis (the "passage"). **Pelvimetry** is the assessment of the pelvic dimensions to determine its adequacy for vaginal delivery. A mismatch between the size of the fetal head and the maternal pelvis is termed **cephalopelvic disproportion (CPD)**. The Pelvic Planes and Their Critical Diameters: The true pelvis can be conceptualized as a curved canal with three key planes, each with a unique shape and set of dimensions. **Pelvic Inlet:** The entrance to the true pelvis is widest in the transverse dimension. **Anteroposterior (AP) Diameter:** The **obstetric conjugate**, measured from the sacral promontory to the posterior surface of the pubic symphysis, is the shortest AP diameter and the most critical measurement of the inlet. Its average value is 10.5–11.0 cm. **Transverse Diameter:**

The widest distance between the iliopectineal lines, averaging 13.0–13.5 cm. **Oblique Diameter:** Measured from one sacroiliac joint to the contralateral iliopubic eminence, averaging 12.0–12.5 cm. **Mid-Pelvis:** This is the plane of least pelvic dimensions, at the level of the ischial spines. **Anteroposterior (AP) Diameter:** From the inferior border of the pubic symphysis to the sacrum, averaging 11.5 cm. **Transverse (Interspinous) Diameter:** The distance between the ischial spines. This is typically the narrowest diameter of the entire pelvis, averaging 10.0–10.5 cm. Midpelvic contraction is suspected if this diameter is less than 10 cm. **Pelvic Outlet:** The exit of the true pelvis is widest in the anteroposterior dimension. **Anteroposterior (AP) Diameter:** From the inferior border of the pubic symphysis to the tip of the sacrum (the coccyx can move posteriorly, increasing this length). It averages 11.5–13.5 cm. **Transverse (Bituberous) Diameter:** The distance between the inner surfaces of the ischial tuberosities, averaging 11.0 cm. **The Cardinal Movements of Labor:** The fetal head, with its own set of diameters (e.g., biparietal diameter of ~9.5 cm), must negotiate the changing dimensions of the pelvic canal. This is achieved through a precise sequence of movements: **Engagement:** The fetal head enters the pelvic inlet. Because the inlet is widest transversely, the head typically engages in an occiput transverse position, aligning its longest axis with the widest pelvic diameter. **Descent:** Continuous downward movement through the pelvis. **Flexion:** As the head descends, it meets resistance from the pelvic floor, causing the neck to flex. This presents the smallest head diameter (suboccipitobregmatic, ~9.5 cm) to the pelvic canal. **Internal Rotation:** This is the most critical movement. As the flexed head reaches the mid-pelvis, the funnel shape of the pelvic floor muscles guides it to rotate 90 degrees, moving from an occiput transverse to an occiput anterior position. This crucial rotation aligns the long axis of the fetal head with the long AP diameter of the pelvic outlet. **Extension:** With the occiput under the pubic symphysis, the head extends, and the brow, nose, mouth, and chin are born over the perineum. **External Rotation (Restitution):** After delivery, the head rotates back to its transverse position to align with the shoulders, which are still negotiating the pelvis. **Expulsion:** The anterior shoulder is delivered under the pubic symphysis, followed by the posterior shoulder and the rest of the body. Failure of these movements to occur, known as arrest of labor, can be caused by CPD or fetal malposition and may require intervention such as Cesarean section.

Pelvic Plane	Diameter Name	Anatomical Boundaries	Average Measurement (cm)	Obstetric Significance
Inlet	Obstetric Conjugate (AP)	Sacral promontory to posterior pubic symphysis	10.5–11.0	Narrowest fixed AP diameter of the pelvis; determines if the head can engage.
Inlet	Transverse	Widest distance between iliopectineal lines	13.0–13.5	Widest diameter of the inlet; dictates transverse engagement of the fetal head.

Pelvic Plane	Diameter Name	Anatomical Boundaries	Average Measurement (cm)	Obstetric Significance
Inlet	Oblique	Sacroiliac joint to contralateral ilipubic eminence	12.0–12.5	Diameter through which the fetal head often engages.
Mid-pelvis	Anteroposterior (AP)	Inferior pubic symphysis to sacrum (at S4-S5)	~11.5	Plane of greatest pelvic dimensions.
Mid-pelvis	Transverse (Interspinous)	Between the tips of the ischial spines	10.0–10.5	Narrowest diameter of the entire pelvis; crucial for mid-pelvic descent.
Outlet	Anteroposterior (AP)	Inferior pubic symphysis to tip of sacrum	11.5–13.5	Widest diameter of the outlet; allows for delivery of the head by extension.
Outlet	Transverse (Bituberous)	Between the inner borders of the ischial tuberosities	~11.0	Determines the space available for the fetal head at the outlet.

Part II

The Female Reproductive System: Anatomy, Histology, and Embryology

The female reproductive system is a dynamic set of organs responsible for oogenesis, fertilization, gestation, parturition, and the production of sex hormones. It consists of internal genitalia located within the true pelvis and external genitalia situated in the perineum.

6 Gross and Applied Anatomy of the Internal and External Genitalia

6.1 Internal Genitalia

Uterus: The uterus is a thick-walled, pear-shaped, hollow muscular organ situated in the lesser pelvis, typically positioned between the urinary bladder anteriorly and the rectum posteriorly. In a nulliparous woman, it measures approximately 3 x 2 x 1 inches. It is composed of four regions: the **fundus**, the dome-shaped portion superior to the entry points of the uterine tubes; the **body (or corpus)**, which is the main central portion; the **isthmus**, a constricted region connecting the body to the cervix; and the **cervix**, the inferior, cylindrical part that projects into the vagina. The uterus is normally **anteverted** (angled anteriorly relative to the axis of the vagina) and **anteflexed** (flexed anteriorly at the level of the isthmus), causing it to rest upon the superior surface of the bladder. Its wall consists of three layers: the **perimetrium**, an outer serous layer derived from the peritoneum; the **myometrium**, a thick middle layer of smooth muscle that provides the contractile force for labor; and the **endometrium**, the inner mucosal lining that undergoes cyclical changes during the menstrual cycle. The primary blood supply is from the **uterine artery**, a branch of the internal iliac artery, which forms extensive anastomoses with the ovarian artery, a direct branch of the abdominal aorta. **Ovaries:** The ovaries are the paired female gonads, almond-shaped organs located on the lateral walls of the pelvis in a depression known as the **ovarian fossa**. They are responsible for producing oocytes (eggs) and secreting the primary female sex hormones, estrogen and progesterone. Each ovary is attached to the posterior layer of the broad ligament by a peritoneal fold called the **mesovarium**. It is connected to the uterus medially by the **ovarian ligament** and to the lateral pelvic wall by the **suspensory ligament of the ovary**, which contains the ovarian artery, vein, nerves, and lymphatics. **Fallopian (Uterine) Tubes:** These are paired, muscular J-shaped tubes, approximately 10-12 cm long, that extend laterally from the superolateral angles (cornua) of the uterus to the vicinity of the ovaries. They serve as the conduit for the oocyte to travel from the ovary to the uterus and are the usual site of fertilization. Each tube is divided into four parts, from lateral to medial: The **infundibulum**, a funnel-shaped opening near the ovary. **Fimbriae**, finger-like projections at the end of the infundibulum that sweep over the ovary to capture the released oocyte. The **ampulla**, the widest and longest part of the tube, where fertilization typically occurs. The **isthmus**, a narrow, thick-walled section that joins the uterus. **Vagina:** The vagina is a fibromuscular tube, approximately 7-9 cm long, that connects the cervix of the uterus to the exterior of the body at the vestibule. It serves as the female organ of copulation, the excretory duct for menstrual flow, and the inferior part of the birth canal. The cervix projects into the anterior wall of the vagina, creating recesses known as the **vaginal fornices** (anterior, posterior, and lateral). The posterior fornix is the deepest and is in close relation to the recto-uterine pouch (of Douglas). The vagina is supported by surrounding pelvic floor muscles and condensations of endopelvic fascia.

Ligament Name	Type	Attachments	Contents	Primary Function
Broad Ligament	Peritoneal Fold	Extends from the lateral margins of the uterus to the lateral pelvic walls and floor.	Uterine tube, ovarian ligament, round ligament, uterine and ovarian vessels, nerves, lymphatics.	Encloses and stabilizes the uterus and its adnexa; serves as a mesentery for neurovasculature.
Round Ligament of Uterus	True Ligament (Fibromuscular)	From the uterine cornu, through the inguinal canal, to the labia majora.	Artery of the round ligament (Sampson's artery).	Maintains the anteverted position of the uterus, especially during pregnancy.
Cardinal (Transverse Cervical) Ligament	True Ligament (Fascial Condensation)	Extends from the cervix and lateral vaginal fornices to the lateral pelvic walls.	Uterine artery and veins.	Provides the primary lateral support for the uterus and cervix (apical support).
Uterosacral Ligament	True Ligament (Fascial Condensation)	Extends from the posterior aspect of the cervix, around the rectum, to the anterior surface of the sacrum.	N/A	Pulls the cervix posteriorly, maintaining the anteverted position of the uterus.
Ovarian Ligament	True Ligament (Fibromuscular)	Connects the medial pole of the ovary to the uterine cornu, posterior to the fallopian tube.	N/A	Tethers the ovary to the uterus.
Suspensory Ligament of the Ovary	Peritoneal Fold	Extends from the superolateral pole of the ovary to the lateral pelvic wall.	Ovarian artery, vein, nerves, and lymphatics.	Carries the neurovascular supply to and from the ovary.

6.2 External Genitalia (Vulva)

The external female genital organs are collectively termed the **vulva**. **Mons Pubis:** A rounded mass of fatty tissue overlying the pubic symphysis, covered with hair after puberty. **Labia Majora:** Two prominent, elongated folds of skin and adipose tissue that form the lateral boundaries of the vulva. They are homologous to the male scrotum. **Labia Minora:** Two smaller, hairless folds of skin located medial to the labia majora. They enclose the vestibule. **Vestibule:** The space enclosed by the labia minora, which contains the external urethral orifice anteriorly and the vaginal orifice (introitus) posteriorly. **Clitoris:** An erectile organ located at the anterior junction of the labia minora. It is homologous to the male penis and is a primary center of sexual sensation. **Vestibular Glands:** The **greater**

vestibular (Bartholin's) glands, located on each side of the vaginal orifice, secrete mucus for lubrication during sexual arousal. The numerous **lesser vestibular (Skene's) glands** open near the urethral orifice.

7 Microscopic Anatomy (Histology) of the Female Reproductive Organs

The microscopic structure of the female reproductive organs is characterized by dynamic changes that are tightly regulated by the cyclical fluctuations of ovarian hormones. **Ovary:** The ovary is composed of an outer **cortex** and an inner **medulla**. The cortex contains the ovarian follicles embedded in a highly cellular stroma. The entire organ is covered by a simple cuboidal layer known as the **germinal epithelium**, which is continuous with the peritoneal mesothelium. Deep to this is a dense connective tissue capsule, the **tunica albuginea**. **Ovarian Folliculogenesis:** This is the process of oocyte maturation within a follicle, driven by pituitary gonadotropins (FSH and LH). **Primordial Follicle:** The most immature stage, consisting of a primary oocyte (arrested in prophase I of meiosis) surrounded by a single layer of flattened follicular (pre-granulosa) cells. **Primary Follicle:** The oocyte enlarges, and the follicular cells become cuboidal and proliferate, forming multiple layers (now called granulosa cells). A glycoprotein coat, the **zona pellucida**, forms around the oocyte. **Secondary (Antral) Follicle:** The granulosa cells secrete follicular fluid, which accumulates in small pockets that coalesce to form a single large cavity, the **antrum**. The surrounding stromal cells organize into the **theca interna** (which produces androgens) and the **theca externa** (a fibrous capsule). **Graafian (Mature) Follicle:** The dominant follicle enlarges significantly. The oocyte, having completed meiosis I to become a secondary oocyte, is perched on a mound of granulosa cells called the **cumulus oophorus**. The granulosa cells immediately surrounding the oocyte form the **corona radiata**. This follicle bulges from the ovarian surface and ruptures at ovulation. **Corpus Luteum and Albicans:** After ovulation, the collapsed follicle transforms into the **corpus luteum**, a temporary endocrine gland composed of granulosa lutein and theca lutein cells, which secretes progesterone and estrogen to support a potential pregnancy. In the absence of pregnancy, it degenerates into a white fibrous scar, the **corpus albicans**. **Uterine (Fallopian) Tube:** The wall has a highly folded mucosa, a muscularis layer, and an outer serosa. The mucosal lining is a simple columnar epithelium with two cell types: **ciliated cells**, whose cilia beat in unison toward the uterus to propel the ovum, and non-ciliated **peg cells**, which secrete a nutritive fluid for the ovum and sperm. **Uterus (Endometrium):** The endometrium is uniquely adapted for implantation and menstruation. It consists of a deep **stratum basalis** and a superficial **stratum functionalis**. The basal layer is retained during menstruation and serves as the source for regeneration of the functional layer, which undergoes profound cyclical changes: **Menstrual Phase (Days 1-4):** Triggered by the decline in progesterone and estrogen, the spiral arteries in the functionalis layer constrict, leading to ischemia, tissue necrosis, and shedding of the entire layer, resulting in menstrual bleeding. **Proliferative Phase (Days 5-14):** Under the influence of rising estrogen levels from the developing ovarian follicles, the functionalis is rebuilt. The epithelium, stroma, and glands proliferate. The glands are straight and tubular with narrow lumens, and spiral arteries begin to lengthen.

Secretory Phase (Days 15-28): Driven by progesterone from the corpus luteum, the endometrium becomes thick, vascular, and edematous in preparation for implantation. The glands become highly coiled, tortuous ("saw-toothed" in appearance), and distended with glycogen-rich secretions. The spiral arteries become long and tightly coiled, extending to the surface of the endometrium. **Cervix:** The cervix has two distinct epithelial linings. The **endocervix** (canal) is lined by a simple columnar, mucus-secreting epithelium that forms deep crypts or glands. The **ectocervix** (vaginal portion) is covered by a durable stratified squamous non-keratinized epithelium. The junction between these two epithelia is the **squamocolumnar junction (SCJ)**. Throughout life, especially during puberty and first pregnancy, the columnar epithelium everts onto the ectocervix and is gradually replaced by squamous epithelium through a process of metaplasia. This dynamic area of change is called the **transformation zone (TZ)** and is the site where most cervical cancers arise, primarily due to its susceptibility to human papillomavirus (HPV) infection. **Vagina:** The vaginal wall has a mucosa lined by stratified squamous non-keratinized epithelium, a muscularis layer, and an outer adventitia. Under the influence of estrogen, the epithelial cells accumulate large amounts of glycogen. When these cells are shed, the glycogen is metabolized by the resident lactobacilli to produce lactic acid, creating an acidic vaginal pH (3.8-4.5) that helps suppress the growth of pathogenic microorganisms. The vaginal wall contains no glands; lubrication is provided by cervical mucus and a plasma transudate from its rich vascular supply during sexual arousal.

8 Clinical Correlates of the Female Reproductive System

Ectopic Pregnancy: This occurs when a fertilized ovum implants outside the uterine cavity. Over 95% of ectopic pregnancies occur in the fallopian tube, most commonly in the ampulla. Conditions that damage the tubal mucosa or impair its motility, such as pelvic inflammatory disease (PID) leading to salpingitis (inflammation of the tubes), increase the risk by trapping the embryo. A ruptured tubal pregnancy is a life-threatening surgical emergency causing massive intra-abdominal hemorrhage. **Uterine Leiomyomas (Fibroids):** These are benign monoclonal tumors arising from the smooth muscle cells (myometrium) of the uterus and are the most common pelvic tumor in women. Their growth is estrogen-dependent, causing them to enlarge during the reproductive years and regress after menopause. They are classified by their location: **submucosal** (projecting into the uterine cavity), **intramural** (within the myometrial wall), and **subserosal** (projecting from the outer surface). Symptoms are highly dependent on location and size and include heavy menstrual bleeding (menorrhagia), often from submucosal fibroids that distort the endometrium; pelvic pressure or pain ("bulk symptoms") from large intramural or subserosal fibroids; and infertility or recurrent pregnancy loss. **Endometriosis:** This is a chronic, inflammatory, estrogen-dependent condition defined by the presence of endometrial-like glands and stroma outside the uterine cavity. The most widely accepted theory of pathogenesis is **retrograde menstruation**, where viable endometrial cells are refluxed through the fallopian tubes into the peritoneal cavity. These ectopic implants most commonly occur on the ovaries, pelvic peritoneum, uterosacral ligaments, and recto-vaginal septum. The tissue responds to cyclical hormones, leading to bleeding, chronic inflammation, fibrosis, and adhesion formation.

This results in the classic symptoms of severe dysmenorrhea (painful periods), chronic pelvic pain, dyspareunia (painful intercourse), and infertility. Endometriosis on an ovary can form a blood-filled cyst known as an endometrioma or "chocolate cyst".

Part III

The Male Reproductive System: Anatomy, Histology, and Embryology

The male reproductive system is comprised of a series of organs responsible for producing, maturing, storing, and transporting sperm, as well as for producing the male sex hormone, testosterone. The system includes both external and internal genitalia.

9 Gross and Applied Anatomy of the Internal and External Genitalia

9.1 External Genitalia

Scrotum: A dual-chambered fibromuscular cutaneous sac located inferior to the penis, containing the testes, epididymides, and the inferior part of the spermatic cords. Its wall contains the **dartos muscle** (smooth muscle that wrinkles the scrotal skin) and the **cremaster muscle** (skeletal muscle that elevates the testes). These muscles work together in the cremasteric reflex to regulate testicular temperature, keeping it slightly below core body temperature, which is essential for normal spermatogenesis. **Penis:** The male organ of copulation, which also serves as the common outlet for urine and semen. It is composed of a **root**, a **body (shaft)**, and the **glans penis**. The body contains three cylindrical columns of erectile tissue: the two dorsal **corpora cavernosa** and the single ventral **corpus spongiosum**, which surrounds the spongy urethra and expands distally to form the glans. The root of the penis is attached to the perineum and is composed of the **bulb** (the proximal end of the corpus spongiosum) and the two **crura** (the proximal ends of the corpora cavernosa).

9.2 Internal Genitalia

Testes: The paired, oval-shaped primary male gonads, or testicles, that reside within the scrotum. Each testis is encased in a tough, fibrous outer capsule, the **tunica albuginea**, which also extends into the testis to form septa that divide it into approximately 250 lobules.

Each lobule contains one to four tightly coiled **seminiferous tubules**, the site of sperm production. The testes produce both sperm (exocrine function) and testosterone (endocrine function). **Epididymis:** A single, long (4-6 meters), highly coiled duct situated on the posterolateral surface of each testis. It is divided into a **head**, which receives immature sperm from the testis via the efferent ductules; a **body**; and a **tail**, where mature sperm are stored before ejaculation. The epididymis is the primary site for sperm maturation, where they acquire motility and the ability to fertilize an egg. **Ductus (Vas) Deferens:** A thick-walled muscular tube, approximately 45 cm long, that is continuous with the tail of the epididymis. It ascends from the scrotum within the spermatic cord, traverses the inguinal canal to enter the pelvic cavity, and courses medially, passing superior to the ureter, to reach the posterior surface of the bladder. Its terminal part dilates to form the **ampulla** before it joins the duct of the seminal vesicle. Its primary function is to rapidly transport sperm from the epididymis toward the urethra during ejaculation via strong peristaltic contractions of its thick muscular wall. **Spermatic Cord:** This cord-like structure suspends the testis in the scrotum and runs between the scrotum and the deep inguinal ring. It is a collection of structures enclosed within layers of fascia derived from the anterior abdominal wall. **Accessory Glands:** These glands produce seminal fluid, which mixes with sperm to form semen. **Seminal Vesicles:** Paired, elongated glands located posterior to the bladder, superior to the prostate. They produce a viscous, alkaline, fructose-rich fluid that constitutes about 70% of the semen volume. This fluid provides energy for the sperm and helps neutralize the acidic environment of the male urethra and female vagina. The duct of each seminal vesicle joins the ampulla of the ductus deferens to form the ejaculatory duct. **Prostate Gland:** A single, walnut-sized fibromuscular gland located inferior to the bladder, encircling the first part of the urethra (the prostatic urethra). It secretes a thin, milky, slightly acidic fluid containing citrate (a nutrient), seminalplasmin (an antibiotic), and various enzymes, including prostate-specific antigen (PSA), which helps liquefy the coagulated semen after ejaculation. **Bulbourethral (Cowper's) Glands:** Paired, pea-sized glands located in the deep perineal pouch. They secrete a thick, clear mucus during sexual arousal that lubricates the urethra and neutralizes any residual acidic urine. **Ejaculatory Ducts:** These are two slender tubes, each formed by the union of the ductus deferens and the duct of the seminal vesicle. They pass through the posterior part of the prostate gland and open into the prostatic urethra on either side of the seminal colliculus.

Category	Structure	Description
Ducts	Ductus (Vas) Deferens	A thick-walled muscular tube that transports sperm from the epididymis.
Arteries	Testicular Artery	A direct branch of the abdominal aorta; the primary arterial supply to the testis and epididymis.
	Artery of the Ductus Deferens	A small artery, usually a branch of the superior or inferior vesical artery, that supplies the ductus deferens.

Category	Structure	Description
	Cremasteric Artery	A branch of the inferior epigastric artery that supplies the cremaster muscle and fascial layers.
Veins	Pampiniform Plexus	An extensive network of up to 12 veins that surround the testicular artery, forming a countercurrent heat exchanger to cool arterial blood before it reaches the testis. It converges to form the testicular vein.
Nerves	Genital branch of the Genitofemoral Nerve	A somatic nerve (L1, L2) that provides motor innervation to the cremaster muscle and sensory innervation to the anterior scrotal skin.
	Sympathetic and Parasympathetic Nerve Fibers	Autonomic nerves that supply the ductus deferens and blood vessels.
Lymphatics	Lymphatic Vessels	Drain lymph from the testis and epididymis to the lumbar (para-aortic) lymph nodes.
Fascial Layers	External Spermatic Fascia	The outermost layer, derived from the external oblique aponeurosis.
	Cremasteric Fascia and Muscle	The middle layer, derived from the internal oblique muscle and its fascia.
	Internal Spermatic Fascia	The innermost layer, derived from the transversalis fascia.

10 Microscopic Anatomy (Histology) of the Male Reproductive Organs

Testis: The parenchyma of the testis is composed of seminiferous tubules and interstitial tissue. **Seminiferous Tubules:** These are the sites of spermatogenesis and are lined by a complex germinal epithelium containing two main cell types. **Spermatogenic Cells:** These are germ cells at various stages of maturation, arranged in layers from the basement membrane to the lumen: spermatogonia (diploid stem cells), primary spermatocytes (undergoing meiosis I), secondary spermatocytes (undergoing meiosis II), and spermatids (haploid cells that undergo spermiogenesis to become spermatozoa). **Sertoli (Sustentacular) Cells:** These are large, non-dividing columnar cells that extend from the basement membrane to the lumen. They are joined by tight junctions, forming the **blood-testis barrier**, which isolates the developing gametes from the immune system. Sertoli cells provide structural and metabolic support to the spermatogenic cells, phagocytose excess cytoplasm from spermatids, and secrete fluids and proteins (like androgen-binding protein) essential for spermatogenesis. **Interstitial Tissue:** The connective tissue stroma between the seminiferous tubules contains clusters of **interstitial cells of Leydig**. These are endocrine cells that produce and secrete testosterone in response to stimulation by luteinizing hormone (LH) from the pituitary gland. **Epididymis and Ductus Deferens:** The long duct of the epididymis and the ductus deferens

are lined by pseudostratified columnar epithelium characterized by cells bearing long, non-motile microvilli called **stereocilia**. These stereocilia are involved in absorbing testicular fluid and secreting substances that contribute to sperm maturation. The ductus deferens is distinguished by its exceptionally thick muscular wall, composed of inner longitudinal, middle circular, and outer longitudinal layers of smooth muscle, which generates powerful peristaltic waves during ejaculation. **Prostate Gland:** The prostate is a collection of 30-50 compound tubuloalveolar glands embedded within a dense fibromuscular stroma. The glandular epithelium is typically simple or pseudostratified columnar. A characteristic feature, particularly in older men, is the presence of laminated prostatic concretions called **corpora amylacea** within the glandular lumens. Histologically, the gland is divided into zones: the **peripheral zone** (where most cancers arise), the **central zone**, and the **transition zone** (where BPH originates). **Seminal Vesicles:** The wall of a seminal vesicle consists of a highly folded mucosa, which creates a complex, labyrinthine or honeycomb-like appearance in cross-section. The lining is a pseudostratified columnar epithelium rich in secretory granules. A thin muscularis layer surrounds the mucosa.

11 Clinical Correlates of the Male Reproductive System

Benign Prostatic Hyperplasia (BPH): BPH is an age-related, non-malignant enlargement of the prostate gland, primarily involving the proliferation of stromal and epithelial cells in the transition zone. As this zone surrounds the prostatic urethra, its enlargement compresses the urethral lumen, leading to bladder outlet obstruction. This manifests clinically with a constellation of lower urinary tract symptoms (LUTS), which can be categorized as **obstructive** (e.g., weak stream, hesitancy, incomplete emptying, terminal dribbling) and **irritative** (e.g., frequency, urgency, nocturia) due to secondary bladder changes like detrusor hypertrophy and instability. Diagnosis is based on clinical symptoms, a digital rectal exam (which may reveal a symmetrically enlarged, smooth, firm gland), and urodynamic studies. **Testicular Torsion:** This is a true urological emergency that occurs when the testis rotates around the vertical axis of the spermatic cord, twisting the neurovascular supply and compromising blood flow. This leads to venous congestion, edema, and subsequent arterial ischemia, which can cause irreversible testicular necrosis within hours if not corrected. The underlying anatomical predisposition is often a "**bell-clapper deformity**," a congenital anomaly where the tunica vaginalis has an abnormally high attachment on the spermatic cord, allowing the testis to hang freely and rotate within the scrotum. The classic clinical presentation is the sudden onset of severe, unilateral scrotal pain, often accompanied by nausea and vomiting. A key physical finding is an absent **cremasteric reflex** on the affected side. **Varicocele:** This condition is defined as an abnormal dilation and tortuosity of the pampiniform plexus of veins within the spermatic cord. It is analogous to a varicose vein in the leg and is often described on physical examination as feeling like a "bag of worms". Varicoceles are far more common on the left side (80-90% of cases) due to a unique anatomical arrangement: the long left testicular vein drains at a right angle into the high-pressure left renal vein, whereas the shorter right testicular vein drains obliquely into the low-pressure inferior vena cava. This makes the left side more susceptible to increased hydrostatic pressure and valvular incompetence.

While often asymptomatic, varicoceles are the most common correctable cause of male infertility. The proposed mechanism for impaired spermatogenesis is an elevation of testicular temperature due to the pooling of warm venous blood, which disrupts the normal testicular thermoregulation.

Part IV

Embryological Development, Homologies, and Congenital Anomalies

The development of the male and female reproductive systems is a remarkable process of differentiation from a common, indifferent embryonic blueprint. This shared origin is the basis for homologous structures and provides a framework for understanding the pathogenesis of various congenital anomalies.

12 Development of the Reproductive Systems: From Indifferent to Differentiated

For the first six weeks of gestation, the reproductive systems of XX and XY embryos are morphologically indistinguishable, a period known as the **indifferent stage**. During this time, three key sets of bipotential structures are established: **Indifferent Gonads**: Primordial germ cells, originating in the yolk sac, migrate to the posterior abdominal wall and populate paired longitudinal ridges of mesoderm called the **gonadal ridges**. This interaction induces the formation of the indifferent gonads, which consist of an outer cortex and an inner medulla. **Paired Genital Ducts**: Two pairs of ducts develop side-by-side: the **mesonephric (Wolffian) ducts**, which are part of the embryonic kidney system, and the **paramesonephric (Müllerian) ducts**, which form adjacent to them. **Indifferent External Genitalia**: The external structures consist of a **genital tubercle**, paired **urogenital folds**, and paired **labioscrotal swellings**. Sexual differentiation begins around the seventh week and is orchestrated by a cascade of genetic and hormonal signals. The pivotal event is the presence or absence of the **SRY gene** (Sex-determining Region on the Y chromosome). **Male Development (SRY Present)**: The SRY gene produces **Testis-Determining Factor (TDF)**, a protein that directs the medullary cords of the indifferent gonad to differentiate into testes. The newly formed testes begin to secrete two crucial hormones around the eighth week: **Anti-Müllerian Hormone (AMH)**, produced by Sertoli cells, causes the regression and disappearance of the Müllerian ducts. **Testosterone**, produced by Leydig cells, stimulates the Wolffian ducts to persist and develop into the male internal reproductive tract: the epididymis, ductus deferens,

and seminal vesicles. In the external genitalia, testosterone is converted locally to the more potent androgen, **dihydrotestosterone (DHT)**. DHT drives the masculinization of the external structures: the genital tubercle elongates to form the penis, the urogenital folds fuse to form the spongy urethra, and the labioscrotal swellings fuse to form the scrotum. **Female Development (SRY Absent):** In the absence of the SRY gene, the cortex of the indifferent gonad develops into an ovary. Without testicular hormones: The absence of AMH allows the Müllerian ducts to persist and develop into the female internal reproductive tract: the fallopian tubes, uterus, and upper portion of the vagina. The absence of testosterone causes the Wolffian ducts to degenerate and disappear. The absence of DHT leads to the development of female external genitalia: the genital tubercle becomes the clitoris, the unfused urogenital folds become the labia minora, and the unfused labioscrotal swellings become the labia majora.

13 Homologous Structures of the Male and Female Reproductive Systems

Because the male and female reproductive systems arise from the same embryonic precursors, their adult structures are considered homologous. Understanding this shared embryological origin provides a unifying framework for genitourinary anatomy, explaining anatomical relationships and clinical phenomena that are otherwise obscure. For example, the course of the round ligament of the uterus through the inguinal canal to the labia majora is explained by its homology with the gubernaculum testis, which guides the descent of the testis into the scrotum. Similarly, the Skene's glands in the female vestibule are considered homologous to the prostate gland, as both arise from the urogenital sinus. This principle demonstrates that the adult male and female forms are not entirely separate designs but rather variations on a single developmental theme, sculpted by the presence or absence of specific hormonal signals during a critical developmental window.

Embryonic Precursor	Adult Female Structure	Adult Male Structure
Indifferent Gonad	Ovary	Testis
Paramesonephric (Müllerian) Duct	Uterine tubes, Uterus, Cervix, Upper Vagina	Appendix testis, Prostatic utricle
Mesonephric (Wolffian) Duct	Epoophoron, Paroophoron, Gartner's duct (vestigial)	Epididymis, Ductus deferens, Seminal vesicle, Ejaculatory duct
Urogenital Sinus	Bladder, Urethra, Skene's glands, Bartholin's glands, Lower Vagina	Bladder, Urethra (prostatic and membranous), Prostate gland, Bulbourethral glands
Genital Tubercle	Clitoris (Glans and Body)	Penis (Glans and Body)
Urogenital Folds	Labia Minora	Ventral aspect of penile shaft, Penile raphe
Labioscrotal Swellings	Labia Majora	Scrotum

Embryonic Precursor	Adult Female Structure	Adult Male Structure
Gubernaculum	Ovarian ligament, Round ligament of the uterus	Gubernaculum testis

14 Common Congenital Anomalies and Their Embryological Basis

Errors in the complex process of reproductive development can lead to a wide range of congenital anomalies. The close embryological relationship between the genital and urinary systems means that an anomaly in one system often warrants investigation of the other. For instance, Müllerian duct anomalies are associated with a high incidence (up to 40%) of concomitant renal abnormalities, such as unilateral renal agenesis, because both systems derive from the intermediate mesoderm and are developmentally intertwined.

14.1 Female Congenital Anomalies

These most often involve the Müllerian ducts and are classified based on the nature of the developmental failure (agenesis, fusion defect, or canalization defect). **Müllerian Agenesis (Mayer-Rokitansky-Kuster-Hauser Syndrome):** Complete or partial absence of the uterus and vagina due to failure of the Müllerian ducts to develop. Ovarian function is typically normal. **Fusion Defects: Uterus Didelphys:** Complete failure of Müllerian duct fusion results in a double uterus, double cervix, and often a double vagina. **Bicornuate Uterus:** Partial failure of fusion at the superior end results in a heart-shaped uterus with two cornua. **Unicornuate Uterus:** Arrested development of one Müllerian duct results in a single-horned, half-sized uterus. **Canalization/Resorption Defects: Septate Uterus:** The most common Müllerian anomaly, resulting from incomplete resorption of the medial septum after the ducts have fused. The septum can be partial or complete, dividing the uterine cavity. **Imperforate Hymen:** Failure of the inferior end of the vaginal plate to canalize, leading to a membrane that completely obstructs the vaginal outlet. At puberty, this causes primary amenorrhea with cyclical pelvic pain due to the accumulation of menstrual blood (hematocolpos).

14.2 Male Congenital Anomalies

Hypospadias: A common anomaly where the external urethral orifice opens on the ventral (under) surface of the penis instead of at the tip of the glans. It results from the incomplete fusion of the urogenital folds. **Cryptorchidism:** Failure of one or both testes to complete their descent from the abdomen into the scrotum. The undescended testis is at increased risk for infertility (due to higher temperature) and testicular cancer.

14.3 Disorders of Sex Development (DSD)

DSDs are conditions where there is a discrepancy between the chromosomal, gonadal, and phenotypic sex. **46,XY DSD (Androgen Insensitivity Syndrome):** Individuals are genetically male (46,XY) but have a defect in the androgen receptor. The SRY gene functions normally, so testes form and produce AMH, causing Müllerian ducts to regress. However, because the body's tissues cannot respond to the testosterone produced by the testes, the Wolffian ducts also regress, and the external genitalia develop along the female pathway. The result is a female phenotype with a blind-ending vagina, no uterus, and undescended testes. **46,XX DSD (Congenital Adrenal Hyperplasia):** Individuals are genetically female (46,XX) and have normal ovaries and internal female ducts. However, an enzymatic defect in the adrenal glands leads to the overproduction of androgens. This excess androgen exposure in utero causes masculinization of the external genitalia, ranging from simple clitoromegaly to the formation of a scrotum-like structure. **46,XY Gonadal Dysgenesis (Swyer Syndrome):** A mutation in the SRY gene prevents the indifferent gonads from developing into testes in a 46,XY individual. Without testes, no AMH or testosterone is produced. Consequently, the Müllerian ducts develop into a uterus and fallopian tubes, and the external genitalia are female. However, the gonads are non-functional "streak" gonads, leading to primary amenorrhea and infertility.

Gross and Clinical Anatomy of the Lower Limb

Introduction

The lower limb is a marvel of biomechanical engineering, designed for the demanding and often opposing functions of providing a stable base for weight-bearing and balance while simultaneously enabling dynamic locomotion. Its architecture reflects these roles, with bones, joints, and muscles that are demonstrably larger and stronger than their counterparts in the upper limb. A thorough understanding of its structure is fundamental for diagnosing and treating the vast array of musculoskeletal, neurological, and vascular conditions that affect this region. This guide provides an exhaustive exploration of the lower limb, organized by both region (gluteal, thigh, leg, and foot) and anatomical system (osteology, arthrology, myology, and neurovasculature). Each section integrates gross anatomical descriptions with relevant clinical correlations, highlighting the direct relationship between structure and function, and between normal anatomy and pathological states.

Part V

The Bony Framework and Articulations

15 Osteology: The Foundation of the Lower Limb

The skeletal architecture of the lower limb provides the fundamental framework for muscle attachment, joint formation, and the transmission of body weight to the ground. An intimate knowledge of its bony landmarks is indispensable for accurate physical examination, interpretation of medical imaging, and the planning of surgical interventions. The design of these bones illustrates a core principle of anatomy: form follows function. The robust construction of the pelvic girdle and the long bones of the leg stands in stark contrast to the lighter, more mobile bones of the upper limb, reflecting a necessary trade-off between stability and dexterity.

15.1 The Pelvic Girdle: The Connection to the Axial Skeleton

The pelvic girdle serves as the crucial link between the axial skeleton and the free lower limb. It is formed by the two hip bones (os coxae or innominate bones), which articulate with the sacrum posteriorly at the sacroiliac joints and with each other anteriorly at the pubic symphysis. This rigid, ring-like structure provides a stable platform for weight transmission. Each adult hip bone is the product of the fusion of three distinct bones during adolescence: the ilium, ischium, and pubis. These three components converge centrally to form the deep, cup-shaped socket of the hip joint, the acetabulum. **Ilium** The ilium is the largest and most superior component of the hip bone, forming the broad, wing-like superior part of the pelvis. Its superior margin, the **iliac crest**, is a prominent, palpable arch that can be felt along the waistline. The iliac crest serves as an attachment site for numerous muscles of the abdomen, back, and lower limb and is a common site for bone marrow biopsy due to its large volume of red marrow. The crest terminates anteriorly at the **anterior superior iliac spine (ASIS)**, a key landmark for identifying the L4 vertebral level and an attachment point for the inguinal ligament and the sartorius muscle. Inferior to the ASIS is a rounded protuberance, the **anterior inferior iliac spine (AIIS)**, which serves as the origin for the straight head of the rectus femoris muscle. Posteriorly, the iliac crest ends at the **posterior superior iliac spine (PSIS)**, which is often visible as a "dimple" on the skin of the lower back. Inferior to this is the **posterior inferior iliac spine (PIIS)**. The large, concave anteromedial surface of the ilium is the **iliac fossa**, which gives rise to the iliacus muscle. The posterior portion of the ilium features the ear-shaped **auricular surface**, which articulates with the sacrum to form the sacroiliac joint, and the large **greater sciatic notch**, an inverted U-shaped indentation that forms the superior border of the greater sciatic foramen. **Ischium** The ischium forms the posteroinferior part of the hip bone. Its most prominent feature is the large, roughened **ischial tuberosity**, which serves as the common origin for the hamstring muscles and supports the body's weight in a seated position. A narrow segment of bone, the

ischial ramus, projects anteriorly to join the inferior pubic ramus. The posterior margin of the ischium features the **ischial spine**, a sharp projection that separates the greater sciatic notch from the smaller **lesser sciatic notch** below it. The ischial spine is a critical landmark in obstetrics and for administering a pudendal nerve block. **Pubis** The pubis is the most anterior component of the hip bone. It consists of a body, a superior pubic ramus, and an inferior pubic ramus. The bodies of the two pubic bones articulate in the midline at the **pubic symphysis**, a cartilaginous joint. The superior pubic ramus extends laterally from the body to meet the ilium, and its superior margin is marked by a sharp ridge known as the **pectineal line**. The inferior pubic ramus extends posterolaterally to join the ischial ramus, forming the **ischiopubic ramus**. **Key Foramina and Notches** The articulated pelvic girdle contains several critical openings that allow for the passage of neurovascular structures and muscles. **Acetabulum:** This is the deep, hemispherical socket on the lateral aspect of the hip bone where the ilium, ischium, and pubis fuse. It articulates with the head of the femur to form the hip joint, providing significant bony congruity and stability. **Obturator Foramen:** This is the large opening located anteroinferior to the acetabulum, bounded by the pubis and ischium. In life, it is almost entirely closed by the fibrous **obturator membrane**, which serves as an attachment site for the obturator internus and externus muscles. A small passageway at its superior aspect, the **obturator canal**, allows the obturator nerve, artery, and vein to pass from the pelvis into the medial compartment of the thigh. **Greater and Lesser Sciatic Foramina:** The greater and lesser sciatic notches of the hip bone are converted into complete foramina by the sacrospinous and sacrotuberous ligaments. The **greater sciatic foramen** is a major gateway from the pelvis to the gluteal region, transmitting the piriformis muscle, the sciatic nerve, the superior and inferior gluteal nerves and vessels, and the pudendal nerve and internal pudendal vessels. The **lesser sciatic foramen** serves as a re-entry point into the perineum for the pudendal nerve and internal pudendal vessels, and as an exit for the tendon of the obturator internus muscle.

15.2 The Femur and Patella: The Thigh and Knee Cap

Femur The femur, or thigh bone, is the longest, heaviest, and strongest bone in the human body, engineered to withstand immense forces during locomotion. **Proximal End:** The proximal femur features the spherical **head**, which is covered with articular cartilage and fits securely into the acetabulum. A small, non-articular pit on the head, the **fovea capitis**, serves as the attachment point for the ligament of the head of the femur. This ligament carries the foveal artery, an important blood supply to the femoral head, particularly in children. The head is connected to the shaft by the constricted **neck**, a common site of fracture in the elderly. Two large prominences serve as major sites for muscle attachment: the large, laterally placed **greater trochanter** and the smaller, posteromedially located **lesser trochanter**. The trochanters are connected anteriorly by the **intertrochanteric line** and posteriorly by the more prominent **intertrochanteric crest**. **Shaft:** The long shaft of the femur is smooth and rounded on its anterior surface but is marked posteriorly by a prominent longitudinal ridge, the **linea aspera**. This ridge is a major site of attachment for the adductor and hamstring muscles. Proximally, the linea aspera diverges into the **pectineal line** medially and the roughened **gluteal tuberosity** laterally. **Distal End:** The

distal femur broadens to form the large **medial and lateral condyles**, which articulate with the tibial plateau to form the tibiofemoral component of the knee joint. Superior to the condyles are the **medial and lateral epicondyles**, which are palpable landmarks and attachment sites for the collateral ligaments of the knee. A small bump on the superior margin of the medial epicondyle, the **adductor tubercle**, marks the insertion of the hamstring portion of the adductor magnus muscle. Posteriorly, the condyles are separated by a deep **intercondylar fossa**, which houses the cruciate ligaments. Anteriorly, the smooth surfaces of the condyles merge to form the **patellar surface**, a groove that articulates with the patella. **Patella** The patella, or kneecap, is the largest sesamoid bone in the body, developing within the tendon of the quadriceps femoris muscle. It is a triangular bone that lies anterior to the knee joint. Its posterior surface articulates with the patellar surface of the femur, forming the patellofemoral joint. By increasing the lever arm of the quadriceps muscle, the patella significantly enhances the efficiency of knee extension.

15.3 The Tibia and Fibula: The Bones of the Leg

The leg contains two bones: the tibia medially and the fibula laterally. **Tibia** The tibia, or shin bone, is the second largest bone in the body and is the primary weight-bearing bone of the leg. **Proximal End:** The proximal end is expanded to form the **medial and lateral condyles**, which together create the flat **tibial plateau**. These surfaces articulate with the femoral condyles. Separating the articular surfaces is the **intercondylar eminence**, an elevated area with two tubercles that serves as the primary attachment site for the anterior and posterior cruciate ligaments. On the anterior surface, just inferior to the condyles, is the prominent **tibial tuberosity**, the insertion point for the patellar ligament. Anterolaterally, a smooth facet known as **Gerdy's tubercle** serves as the insertion point for the iliotibial tract. **Shaft:** The shaft of the tibia is triangular in cross-section. Its sharp anterior border is located just deep to the skin and is easily palpable along its entire length, making it vulnerable to injury. The posterior surface is marked in its upper third by an oblique ridge, the **soleal line**, for the origin of the soleus muscle. **Distal End:** The distal end of the tibia is smaller than the proximal end and features a prominent medial projection, the **medial malleolus**, which forms the medial prominence of the ankle. It articulates with the talus and provides attachment for the deltoid ligament. **Fibula** The fibula is a long, slender bone located lateral to the tibia. It does not bear weight but is essential for muscle attachment and for forming the lateral part of the ankle joint. **Proximal End:** The **head of the fibula** articulates with the lateral condyle of the tibia at the proximal tibiofibular joint. It serves as the attachment site for the biceps femoris tendon and the lateral collateral ligament of the knee. The common fibular nerve courses superficially around the neck of the fibula, just distal to the head, making it susceptible to compression or injury in this location. **Distal End:** The distal end of the fibula forms the **lateral malleolus**, which is more prominent and extends more distally than the medial malleolus. It articulates with the lateral aspect of the talus and provides attachment for the lateral ligaments of the ankle. The shafts of the tibia and fibula are connected by a strong, fibrous **interosseous membrane**, which provides a surface for muscle origin and separates the anterior and posterior compartments of the leg. Both bones possess **nutrient foramina**, small openings through which nutrient arteries enter to

supply the medullary cavity. In the tibia and fibula, these foramina are typically located on the posterior surface of the diaphysis and are directed away from the growing end of the bone (the knee). The architecture of the fibula and talus demonstrates that a bone's clinical importance is not solely defined by its role in weight-bearing. The fibula, while non-weight-bearing, is indispensable for ankle stability. Its lateral malleolus forms the lateral buttress of the ankle mortise, preventing talar shift. Fractures of the fibula, particularly those associated with disruption of the syndesmosis (e.g., a Maisonneuve fracture), can lead to profound ankle instability. Furthermore, its neck is a site of clinical vulnerability for the common fibular nerve, where compression can lead to foot drop. Similarly, the talus bears no direct muscle attachments but acts as a critical link, distributing forces from the tibia to the rest of the foot. Its precarious blood supply makes it highly susceptible to avascular necrosis following fracture, a complication with devastating functional consequences.

15.4 The Bones of the Foot: Tarsals, Metatarsals, and Phalanges

The skeleton of the foot is complex, containing 26 bones designed to provide a stable yet adaptable platform for supporting the body and propelling it forward. The bones are organized into three groups: the tarsals, metatarsals, and phalanges. **Tarsals** The seven tarsal bones form the posterior half of the foot (the ankle and heel region). **Talus:** Sits superiorly, articulating with the tibia and fibula at the ankle joint and with the calcaneus below at the subtalar joint. It is unique in that it has no direct muscular attachments; its movement is produced by muscles acting on other bones. **Calcaneus:** The largest tarsal bone, forming the heel. Its posterior surface has a large tuberosity for the attachment of the powerful calcaneal (Achilles) tendon. **Navicular:** A boat-shaped bone located on the medial side of the foot, articulating with the talus proximally and the three cuneiforms distally. **Cuboid:** A cube-shaped bone on the lateral side of the foot, articulating with the calcaneus proximally and the fourth and fifth metatarsals distally. **Cuneiforms (3):** The medial, intermediate, and lateral cuneiforms are wedge-shaped bones that articulate with the navicular proximally and the first three metatarsals distally. **Metatarsals** The five metatarsal bones form the anterior half of the foot, connecting the tarsals to the phalanges. They are numbered 1 to 5, from the great toe (medial) to the little toe (lateral). The base of the fifth metatarsal has a prominent tuberosity on its lateral side, which serves as the insertion point for the fibularis brevis tendon and is a common site for avulsion fractures. **Phalanges** The 14 phalanges are the bones of the toes. The great toe (hallux) has two phalanges: a proximal and a distal phalanx. The other four toes each have three phalanges: proximal, middle, and distal.

16 Arthrology: The Joints of the Lower Limb

The joints of the lower limb are engineered to provide the stability required for weight-bearing while permitting the wide range of motion necessary for complex activities like walking, running, and climbing. The design of each major joint represents a unique solution to the stability-mobility paradox.

16.1 The Hip Joint (Coxal Joint)

The hip joint is a multiaxial ball-and-socket synovial joint renowned for its high degree of stability, a feature essential for its weight-bearing role. **Articulating Surfaces:** The joint is formed by the articulation of the spherical head of the femur with the deep, cup-shaped acetabulum of the hip bone. The acetabulum's depth is enhanced by a fibrocartilaginous rim called the **acetabular labrum**, which further grasps the femoral head, increasing joint stability and creating a fluid seal. **Stabilizing Structures:** The hip joint's remarkable stability is a product of both its bony architecture and its powerful ligamentous and muscular reinforcements. **Capsular Ligaments:** The fibrous joint capsule is exceptionally strong and is reinforced by three major ligaments that spiral around the joint. The **iliofemoral ligament (Y-ligament of Bigelow)** is an inverted Y-shaped ligament located anteriorly; it is the strongest ligament in the body and prevents hyperextension of the hip during standing. The **pubofemoral ligament** is located anteroinferiorly and limits abduction and extension. The **ischiofemoral ligament** is located posteriorly and limits internal rotation and extension. **Intracapsular Ligament:** The **ligament of the head of the femur (ligamentum teres)** is a relatively weak, intracapsular ligament that connects the fovea capitis on the femoral head to the acetabular notch. It encloses a small artery that contributes to the blood supply of the femoral head, particularly in children.

16.2 The Knee Joint

The knee is the largest and one of the most complex joints in the body. It is functionally a modified hinge joint, allowing primarily for flexion and extension, with a small degree of rotation when the knee is flexed. Its stability relies less on bony congruity and more on its complex array of ligaments and muscles. The knee comprises three articulations within a single synovial cavity: two tibiofemoral joints (between the femoral and tibial condyles) and one patellofemoral joint. **Menisci:** The **medial and lateral menisci** are C-shaped pads of fibrocartilage that sit on the tibial plateau. They serve several critical functions: they deepen the articular surfaces to improve joint congruency, act as shock absorbers by distributing forces over a larger area, and contribute to joint lubrication and stability. The **medial meniscus** is larger, more C-shaped, and is firmly attached to the joint capsule and the medial collateral ligament. This lack of mobility makes it significantly more susceptible to injury than the more mobile, nearly circular **lateral meniscus**. **Extracapsular Ligaments:** These ligaments are located outside the joint capsule and provide reinforcement. **Patellar Ligament:** A strong, flat band that is the continuation of the quadriceps femoris tendon, extending from the apex of the patella to the tibial tuberosity. **Medial (Tibial) Collateral Ligament (MCL):** A broad, flat ligament on the medial side of the knee that connects the medial epicondyle of the femur to the medial condyle of the tibia. It is the primary restraint against valgus forces (forces that push the knee inward). **Lateral (Fibular) Collateral Ligament (LCL):** A strong, cord-like ligament on the lateral side of the knee, extending from the lateral epicondyle of the femur to the head of the fibula. It is the primary restraint against varus forces (forces that push the knee outward). **Intracapsular Ligaments (Cruciate Ligaments):** Named for the fact that they cross each other within the joint ("cruciate" means cross-shaped),

these two ligaments are the primary stabilizers against anteroposterior displacement. **Anterior Cruciate Ligament (ACL):** Arises from the anterior intercondylar area of the tibia and ascends posterolaterally to attach to the medial surface of the lateral femoral condyle. The ACL is the primary restraint against anterior translation of the tibia relative to the femur and also resists rotational forces. **Posterior Cruciate Ligament (PCL):** Arises from the posterior intercondylar area of the tibia and ascends anteromedially to attach to the lateral surface of the medial femoral condyle. The PCL is stronger than the ACL and is the primary restraint against posterior translation of the tibia relative to the femur. **Bursae:** The knee joint is surrounded by more than a dozen bursae, which are fluid-filled sacs that reduce friction between moving structures. Clinically important bursae include the **suprapatellar bursa**, **prepatellar bursa** ("housemaid's knee"), and the **infrapatellar bursae** ("clergyman's knee").

The common clinical entity known as the "unhappy triad" or "terrible triad" of the knee—involving injury to the ACL, MCL, and medial meniscus—is a direct consequence of the knee's interconnected anatomy. A typical mechanism is a valgus force applied to the lateral side of a planted, flexed knee, often with an external rotation component. This force directly stretches the MCL to its breaking point. Because the deep fibers of the MCL are firmly attached to the periphery of the medial meniscus, the same force can pull the meniscus apart, causing a tear. The rotational component of this violent event places extreme tension on the ACL, leading to its rupture. Therefore, a clinician diagnosing a complete MCL tear must maintain a high index of suspicion for these associated injuries, as they stem from a single traumatic event that exploits the structural linkages within the joint.

16.3 The Ankle and Subtalar Joints: Adaptation and Support

Ankle Joint (Talocrural Joint) The ankle joint is a synovial hinge joint responsible for the movements of dorsiflexion and plantarflexion. **Articulating Surfaces:** It is formed by the articulation of the distal ends of the tibia and fibula with the superior surface (trochlea) of the talus. The medial malleolus of the tibia and the lateral malleolus of the fibula form a deep socket, or **mortise**, that grips the talus. The trochlea of the talus is wider anteriorly than posteriorly, which means the joint is most stable in the dorsiflexed position when the wider part of the talus is securely held in the mortise. **Ligaments:** The ankle joint is stabilized by strong medial and lateral ligament complexes. **Medial (Deltoid) Ligament:** This is a very strong, fan-shaped complex of four ligaments that attaches the medial malleolus to the talus, calcaneus, and navicular bones. It is a powerful stabilizer that resists over-eversion of the foot. **Lateral Ligament Complex:** This complex is weaker than the deltoid ligament and consists of three separate ligaments that attach the lateral malleolus to the talus and calcaneus: the **anterior talofibular ligament (ATFL)**, the **posterior talofibular ligament (PTFL)**, and the **calcaneofibular ligament (CFL)**. These ligaments collectively resist over-inversion of the foot. The ATFL is the weakest of the three and is the most commonly injured ligament in an ankle sprain. The overwhelming prevalence of lateral ankle sprains (inversion injuries) is a direct consequence of the ankle's asymmetrical anatomy. The lateral malleolus extends more distally than the medial malleolus, providing a strong bony block that limits eversion. Concurrently, the medial deltoid ligament is a robust, multi-banded structure that

provides formidable resistance to eversion forces. In contrast, the lateral side has less bony restraint against inversion, and its ligamentous support is provided by three separate, weaker ligaments. This inherent anatomical imbalance predisposes the ankle to "roll inward," explaining why approximately 90% of ankle sprains are inversion injuries and why the ATFL is the first and most frequent casualty.

Subtalar Joint (Talocalcaneal Joint) The subtalar joint is a plane synovial joint formed by the articulation between the inferior surface of the talus and the superior surface of the calcaneus. Its primary function is to allow the crucial movements of inversion and eversion of the foot, which are essential for adapting to uneven surfaces during walking and running. The joint is stabilized by a joint capsule and several talocalcaneal ligaments, including the strong **interosseous talocalcaneal ligament** located within the tarsal sinus.

16.4 The Arches of the Foot: Engineering for Weight Distribution

The bones of the foot are not arranged in a flat plane but form three structural arches that provide flexibility, shock absorption, and efficient weight distribution during gait. **Medial Longitudinal Arch:** The highest and most important arch, formed by the calcaneus, talus, navicular, three cuneiforms, and the first three metatarsals. **Lateral Longitudinal Arch:** A lower, flatter arch formed by the calcaneus, cuboid, and the fourth and fifth metatarsals. **Transverse Arch:** Runs from side to side, formed by the cuboid, cuneiforms, and the bases of the metatarsals. The integrity of these arches is maintained by the interlocking shapes of the bones, strong plantar ligaments (such as the plantar calcaneonavicular or "spring" ligament and the long and short plantar ligaments), and the dynamic support provided by the tendons of extrinsic muscles (e.g., tibialis posterior, fibularis longus) and the intrinsic muscles of the foot.

Joint	Ligament Name	Attachments (Proximal to Distal)	Primary Function/Movement Resisted
Hip	Iliofemoral Ligament	Anterior inferior iliac spine (AIIS) to intertrochanteric line	Prevents hyperextension; strongest ligament in the body
	Pubofemoral Ligament	Superior pubic ramus to intertrochanteric line	Prevents excessive abduction and extension
	Ischiofemoral Ligament	Ischium (posterior to acetabulum) to greater trochanter	Prevents excessive internal rotation and extension
	Ligament of the Head of Femur	Acetabular notch to fovea capitis of femoral head	Carries foveal artery to femoral head (especially in children)
Knee	Medial Collateral (MCL)	Medial epicondyle of femur to medial condyle of tibia	Resists valgus (inward) forces
	Lateral Collateral (LCL)	Lateral epicondyle of femur to head of fibula	Resists varus (outward) forces

Joint	Ligament Name	Attachments (Proximal to Distal)	Primary Function/Movement Resisted
	Anterior Cruciate (ACL)	Anterior intercondylar area of tibia to medial side of lateral femoral condyle	Prevents anterior translation of tibia on femur
	Posterior Cruciate (PCL)	Posterior intercondylar area of tibia to lateral side of medial femoral condyle	Prevents posterior translation of tibia on femur
Ankle	Medial (Deltoid) Ligament	Medial malleolus to talus, calcaneus, and navicular	Resists eversion of the foot
	Anterior Talofibular (ATFL)	Lateral malleolus to neck of talus	Resists inversion in plantarflexion; most commonly sprained
	Calcaneofibular (CFL)	Lateral malleolus to lateral surface of calcaneus	Resists inversion in neutral/dorsiflexion
	Posterior Talofibular (PTFL)	Lateral malleolus to posterior tubercle of talus	Resists posterior displacement of talus; strongest lateral ligament

Part VI

The Muscular System and Functional Compartments

The musculature of the lower limb is organized into distinct compartments by deep fascial septa. This compartmentalization is of profound clinical importance, as muscles within a given compartment typically share a common innervation, blood supply, and primary function. Understanding this organization is key to diagnosing nerve injuries and understanding conditions like compartment syndrome.

17 Myology of the Hip and Thigh

17.1 The Gluteal Region: Extensors, Abductors, and Rotators

The muscles of the gluteal region are organized into a superficial group of large, powerful muscles and a deep group of smaller muscles. They are primarily responsible for extending, abducting, and rotating the thigh at the hip joint. **Superficial Group** This group includes the three gluteal muscles and the tensor fasciae latae. The **gluteus maximus** is the largest and most superficial muscle, forming the

bulk of the buttock. It is a powerful extensor of the thigh, essential for rising from a seated position, climbing stairs, and running. It also acts as a lateral rotator of the thigh. It receives its innervation from the inferior gluteal nerve. The **gluteus medius** and **gluteus minimus** lie deep to the gluteus maximus. They are the primary abductors of the thigh and also act as medial rotators. Their most critical function is to stabilize the pelvis during the stance phase of gait, preventing the pelvis from dropping on the unsupported side. Both are innervated by the superior gluteal nerve. The **tensor fasciae latae (TFL)** is a small muscle on the anterolateral aspect of the hip that tenses the iliotibial tract, assisting in abduction and medial rotation of the thigh. It is also innervated by the superior gluteal nerve. **Deep Group (Lateral Rotators)** This group consists of six smaller muscles located deep to the gluteus minimus: the **piriformis**, **obturator internus**, **superior gemellus**, **inferior gemellus**, **quadratus femoris**, and **obturator externus**. As their group name implies, their primary action is lateral rotation of the thigh. They also play a role in stabilizing the hip joint by pulling the femoral head into the acetabulum. The piriformis is a key anatomical landmark, as the sciatic nerve and other neurovascular structures typically emerge from the greater sciatic foramen inferior to its belly.

Muscle	Origin	Insertion	Primary Action(s)	Innervation (Root Value)
Superficial Group				
Gluteus Maximus	Ilium (posterior to posterior gluteal line), dorsal sacrum, coccyx, sacrotuberous ligament	Iliotibial tract, gluteal tuberosity of femur	Extends and laterally rotates thigh; stabilizes trunk	Inferior Gluteal Nerve (L5, S1, S2)
Gluteus Medius	Ilium (between anterior and posterior gluteal lines)	Lateral surface of greater trochanter	Abducts and medially rotates thigh; stabilizes pelvis	Superior Gluteal Nerve (L4, L5, S1)
Gluteus Minimus	Ilium (between anterior and inferior gluteal lines)	Anterior surface of greater trochanter	Abducts and medially rotates thigh; stabilizes pelvis	Superior Gluteal Nerve (L4, L5, S1)
Tensor Fasciae Latae	Anterior superior iliac spine (ASIS), anterior iliac crest	Iliotibial tract	Abducts, medially rotates, and flexes thigh	Superior Gluteal Nerve (L4, L5, S1)
Deep Group (Lateral Rotators)				
Piriformis	Anterior surface of sacrum	Superior border of greater trochanter	Laterally rotates and abducts thigh	Nerve to Piriformis (S1, S2)
Obturator Internus	Internal surface of obturator membrane and surrounding bone	Medial surface of greater trochanter	Laterally rotates and abducts thigh	Nerve to Obturator Internus (L5, S1)

Muscle	Origin	Insertion	Primary Action(s)	Innervation (Root Value)
Superior Gemellus	Ischial spine	Medial surface of greater trochanter (with obturator internus tendon)	Laterally rotates and abducts thigh	Nerve to Obturator Internus (L5, S1)
Inferior Gemellus	Ischial tuberosity	Medial surface of greater trochanter (with obturator internus tendon)	Laterally rotates and abducts thigh	Nerve to Quadratus Femoris (L5, S1)
Quadratus Femoris	Lateral border of ischial tuberosity	Quadratus tubercle on intertrochanteric crest	Laterally rotates thigh	Nerve to Quadratus Femoris (L5, S1)

17.2 The Anterior Thigh: The Quadriceps Femoris Group

The muscles of the anterior compartment of the thigh are primarily responsible for extending the leg at the knee joint and flexing the thigh at the hip joint. They are innervated by the **femoral nerve (L2-L4)**. **Quadriceps Femoris:** This is the large, powerful muscle group that forms the main bulk of the anterior thigh. It consists of four distinct heads that converge to form the common quadriceps tendon, which inserts onto the patella. This force is then transmitted via the patellar ligament to the tibial tuberosity. **Rectus Femoris:** Unique among the quadriceps, it originates from the AIIS of the ilium, allowing it to cross both the hip and knee joints. Consequently, it is the only head that both extends the knee and flexes the thigh. **Vastus Lateralis, Vastus Medialis, and Vastus Intermedius:** These three muscles originate from the shaft of the femur and act solely to extend the knee. **Sartorius:** This is the longest muscle in the body, a strap-like muscle that crosses the thigh obliquely from superolateral to inferomedial. It originates from the ASIS and inserts onto the medial surface of the proximal tibia as part of the **pes anserinus**. It produces a combination of movements: flexion, abduction, and lateral rotation of the thigh, as well as flexion of the knee. **Iliopsoas:** This is the most powerful flexor of the hip joint. It is a composite muscle formed by the **iliacus**, which arises from the iliac fossa, and the **psoas major**, which arises from the lumbar vertebrae. They share a common tendon that inserts on the lesser trochanter of the femur. The iliacus is innervated by the femoral nerve, while the psoas major is innervated by direct branches from the lumbar plexus (L1-L3). **Pectineus:** This is a flat, quadrangular muscle located at the superior aspect of the medial thigh. It acts to adduct and flex the thigh. It is often considered a transitional muscle as it typically receives dual innervation from both the femoral nerve and the obturator nerve.

Muscle	Origin	Insertion	Primary Action(s)	Innervation (Root Value)
Quadriceps Femoris				
Rectus Femoris	Anterior inferior iliac spine (AIIS), supra-acetabular groove	Patella (via quadriceps tendon) and tibial tuberosity (via patellar ligament)	Extends leg; flexes thigh	Femoral Nerve (L2, L3, L4)
Vastus Lateralis	Greater trochanter, lateral lip of linea aspera	Patella (via quadriceps tendon) and tibial tuberosity (via patellar ligament)	Extends leg	Femoral Nerve (L2, L3, L4)
Vastus Medialis	Intertrochanteric line, medial lip of linea aspera	Patella (via quadriceps tendon) and tibial tuberosity (via patellar ligament)	Extends leg	Femoral Nerve (L2, L3, L4)
Vastus Intermedius	Anterior and lateral surfaces of femoral shaft	Patella (via quadriceps tendon) and tibial tuberosity (via patellar ligament)	Extends leg	Femoral Nerve (L2, L3, L4)
Other Muscles				
Sartorius	Anterior superior iliac spine (ASIS)	Superior medial surface of tibia (pes anserinus)	Flexes, abducts, laterally rotates thigh; flexes leg	Femoral Nerve (L2, L3, L4)
Iliopsoas (Iliacus & Psoas Major)	Iliac fossa (iliacus); lumbar vertebrae (psoas major)	Lesser trochanter of femur	Flexes thigh	Femoral Nerve (Iliacus); Anterior Rami L1-L3 (Psoas)
Pectineus	Pectineal line of pubis	Pectineal line of femur	Adducts and flexes thigh	Femoral Nerve (L2, L3); may receive branch from Obturator Nerve

17.3 The Medial Thigh: The Adductor Group

The muscles of the medial compartment of the thigh are primarily responsible for adduction of the thigh at the hip joint. With one key exception, they are all innervated by the **obturator nerve (L2-L4)**. **Adductor Longus, Brevis, and Magnus:** This is a group of three large muscles that originate from the pubis and ischium and insert broadly along the linea aspera of the femur. The **adductor magnus** is the largest and most complex of the group. It has two distinct parts with different innervations, reflecting its

dual developmental origin. The anterior **adductor portion** is innervated by the obturator nerve and adducts and flexes the thigh. The posterior **hamstring portion** originates from the ischial tuberosity, inserts on the adductor tubercle of the femur, and is innervated by the tibial division of the sciatic nerve (L4). This part acts to adduct and extend the thigh, functioning similarly to the hamstring muscles. **Gracilis:** This is a long, thin, strap-like muscle, the most superficial in the medial compartment. It originates from the pubis and descends to insert on the medial surface of the tibia as part of the **pes anserinus**. It is the only muscle of this group to cross the knee joint, and it acts to adduct the thigh and flex the leg. **Obturator Externus:** A small, flat muscle located deep in the superior part of the compartment. It originates from the external surface of the obturator membrane and acts to laterally rotate the thigh. The dual innervation of muscles like the adductor magnus and pectineus is not an anatomical curiosity but rather a reflection of their complex embryological development and multifaceted function. During development, myoblasts migrate from different primitive muscle masses, carrying their nerve supply with them. The adductor magnus forms from the fusion of two such masses, one destined to be an adductor (obturator nerve) and one a hip extensor (tibial nerve). This explains its ability to perform both actions. Clinically, this anatomical arrangement has important implications. For instance, an injury to the obturator nerve will cause significant weakness in thigh adduction, but it will not cause complete paralysis of the movement. The hamstring portion of the adductor magnus (innervated by the tibial nerve) and the pectineus (innervated primarily by the femoral nerve) can still contribute to adduction, a nuance essential for accurate neurological assessment.

Muscle	Origin	Insertion	Primary Action(s)	Innervation (Root Value)
Adductor Longus	Body of pubis	Middle third of linea aspera	Adducts thigh	Obturator Nerve (L2, L3, L4)
Adductor Brevis	Body and inferior ramus of pubis	Pectineal line, proximal linea aspera	Adducts thigh	Obturator Nerve (L2, L3, L4)
Adductor Magnus	Adductor Part: Inferior pubic ramus, ischial ramus Hamstring Part: Ischial tuberosity	Adductor Part: Linea aspera, medial supracondylar line Hamstring Part: Adductor tubercle of femur	Adductor Part: Adducts and flexes thigh Hamstring Part: Adducts and extends thigh	Adductor Part: Obturator Nerve (L2-L4) Hamstring Part: Tibial division of Sciatic Nerve (L4)
Gracilis	Body and inferior ramus of pubis	Superior medial surface of tibia (pes anserinus)	Adducts thigh; flexes leg	Obturator Nerve (L2, L3, L4)
Obturator Externus	Margins of obturator foramen, obturator membrane	Trochanteric fossa of femur	Laterally rotates thigh	Obturator Nerve (L2, L3, L4)

17.4 The Posterior Thigh: The Hamstring Group

The muscles of the posterior compartment of the thigh are collectively known as the hamstrings. They are characterized by their common origin from the ischial tuberosity, their action of extending the thigh and flexing the leg, and their innervation by the tibial division of the sciatic nerve. **Biceps Femoris:** This is the lateral muscle of the group and has two heads. The **long head** arises from the ischial tuberosity and is a true hamstring. The **short head** arises from the linea aspera of the femur and therefore only acts to flex the knee. Uniquely, the short head is innervated by the common fibular division of the sciatic nerve. The two heads unite to form a common tendon that inserts on the head of the fibula. **Semitendinosus:** This muscle lies superficially on the medial side of the posterior thigh. It has a long, cord-like tendon that inserts onto the medial surface of the tibia as part of the **pes anserinus**. **Semimembranosus:** This muscle is broad and flat and lies deep to the semitendinosus on the medial side. It inserts on the posterior aspect of the medial tibial condyle.

Muscle	Origin	Insertion	Primary Action(s)	Innervation (Root Value)
Biceps Femoris	Long Head: Ischial tuberosity Short Head: Linea aspera of femur	Head of fibula	Flexes leg; extends thigh; laterally rotates leg and thigh	Long Head: Tibial division of Sciatic Nerve (L5, S1, S2) Short Head: Common Fibular division of Sciatic Nerve (L5, S1, S2)
Semitendinosus	Ischial tuberosity	Superior medial surface of tibia (pes anserinus)	Flexes leg; extends thigh; medially rotates leg and thigh	Tibial division of Sciatic Nerve (L5, S1, S2)
Semimembranosus	Ischial tuberosity	Posterior surface of medial tibial condyle	Flexes leg; extends thigh; medially rotates leg and thigh	Tibial division of Sciatic Nerve (L5, S1, S2)

18 Myology of the Leg and Foot

18.1 The Anterior and Lateral Compartments of the Leg

Anterior Compartment The four muscles of the anterior compartment of the leg are primarily responsible for dorsiflexion of the ankle and extension of the toes. They are all innervated by the **deep fibular (peroneal) nerve (L4, L5)** and supplied by the anterior tibial artery. **Tibialis Anterior:** The most medial and superficial muscle, it is a powerful dorsiflexor and invertor of the foot. **Extensor Hallucis Longus:** Located deep to the tibialis anterior and extensor digitorum longus, its primary action is to extend the great toe (hallux), but it also assists in dorsiflexion. **Extensor Digitorum Longus:** The most lateral of the deep muscles, it extends the lateral four toes and assists in dorsiflexion. **Fibularis (Peroneus) Tertius:** A small muscle that is often considered a part of the extensor digitorum longus. It is a weak dorsiflexor and an evertor of the foot.

Muscle	Origin	Insertion	Primary Action(s)	Innervation (Root Value)
Tibialis Anterior	Lateral condyle and superior lateral surface of tibia; interosseous membrane	Medial cuneiform, base of 1st metatarsal	Dorsiflexes and inverts foot	Deep Fibular Nerve (L4, L5)
Extensor Hallucis Longus	Middle anterior surface of fibula; interosseous membrane	Dorsal aspect of base of distal phalanx of great toe	Extends great toe; dorsiflexes foot	Deep Fibular Nerve (L5, S1)
Extensor Digitorum Longus	Lateral condyle of tibia; superior anterior surface of fibula; interosseous membrane	Dorsal aspect of middle and distal phalanges of lateral four toes	Extends lateral four toes; dorsiflexes foot	Deep Fibular Nerve (L5, S1)
Fibularis (Peroneus) Tertius	Distal anterior surface of fibula; interosseous membrane	Dorsum of base of 5th metatarsal	Dorsiflexes and everts foot	Deep Fibular Nerve (L5, S1)

Lateral Compartment The lateral compartment of the leg contains two muscles that are the primary evertors of the foot. They also assist in plantarflexion. Both are innervated by the **superficial fibular (peroneal) nerve (L5, S1, S2)**. **Fibularis (Peroneus) Longus:** The more superficial of the two, it originates from the head and superior lateral surface of the fibula. Its long tendon passes posterior to the lateral malleolus, then crosses the sole of the foot to insert on the medial cuneiform and the base of the first metatarsal.

This course allows it to evert the foot and support the transverse and longitudinal arches. **Fibularis (Peroneus) Brevis:** Lies deep to the longus, originating from the inferior lateral surface of the fibula. Its tendon also passes posterior to the lateral malleolus but inserts on the tuberosity at the base of the fifth metatarsal.

Muscle	Origin	Insertion	Primary Action(s)	Innervation (Root Value)
Fibularis (Peroneus) Longus	Head and superior two-thirds of lateral surface of fibula	Base of 1st metatarsal, medial cuneiform	Everts and weakly plantarflexes foot; supports arches	Superficial Fibular Nerve (L5, S1, S2)
Fibularis (Peroneus) Brevis	Inferior two-thirds of lateral surface of fibula	Tuberosity of base of 5th metatarsal	Everts and weakly plantarflexes foot	Superficial Fibular Nerve (L5, S1, S2)

18.2 The Posterior Compartment of the Leg

The posterior compartment contains seven muscles responsible for plantarflexion of the ankle, inversion of the foot, and flexion of the toes. All are innervated by the **tibial nerve (S1, S2)**. The compartment is divided by a deep transverse fascial septum into superficial and deep groups. **Superficial Group** This group, collectively known as the triceps surae, forms the bulk of the calf and inserts via the common **calcaneal (Achilles) tendon** onto the posterior surface of the calcaneus. **Gastrocnemius:** The most superficial muscle, it has two heads (medial and lateral) that arise from the femoral condyles. It is a powerful plantarflexor, especially when the knee is extended, and also flexes the knee. **Soleus:** A broad, flat muscle located deep to the gastrocnemius. It originates from the posterior tibia (soleal line) and fibula. It is the workhorse of plantarflexion, acting regardless of knee position. **Plantaris:** A small, weak muscle with a long, thin tendon. It is absent in about 10% of the population.

Muscle	Origin	Insertion	Primary Action(s)	Innervation (Root Value)
Gastrocnemius	Medial and lateral condyles of femur	Posterior surface of calcaneus via calcaneal tendon	Plantarflexes foot; flexes knee	Tibial Nerve (S1, S2)
Soleus	Soleal line of tibia, posterior head of fibula	Posterior surface of calcaneus via calcaneal tendon	Plantarflexes foot	Tibial Nerve (S1, S2)

Muscle	Origin	Insertion	Primary Action(s)	Innervation (Root Value)
Plantaris	Lateral supracondylar line of femur	Posterior surface of calcaneus (medial to calcaneal tendon)	Weakly assists gastrocnemius	Tibial Nerve (S1, S2)

Deep Group These four muscles lie deep to the triceps surae. **Popliteus:** A thin, triangular muscle that forms the floor of the popliteal fossa. It acts to "unlock" the fully extended knee by initiating flexion via lateral rotation of the femur on the fixed tibia. **Flexor Hallucis Longus (FHL):** Located laterally in the deep compartment, it originates from the fibula. Its tendon runs medially to insert on the distal phalanx of the great toe, which it powerfully flexes. **Flexor Digitorum Longus (FDL):** Located medially, it originates from the tibia. Its tendon divides to insert on the distal phalanges of the lateral four toes, which it flexes. **Tibialis Posterior:** The deepest muscle of the group, situated between the FHL and FDL. It originates from the tibia, fibula, and interosseous membrane. Its tendon passes behind the medial malleolus to insert on several tarsal and metatarsal bones. It is the primary invertor of the foot and a key dynamic supporter of the medial longitudinal arch.

Muscle	Origin	Insertion	Primary Action(s)	Innervation (Root Value)
Popliteus	Lateral condyle of femur, lateral meniscus	Posterior surface of proximal tibia	Unlocks knee (laterally rotates femur on tibia); weakly flexes knee	Tibial Nerve (L4, L5, S1)
Flexor Hallucis Longus	Inferior two-thirds of posterior fibula	Plantar surface of base of distal phalanx of great toe	Flexes great toe; weakly plantarflexes foot	Tibial Nerve (S2, S3)
Flexor Digitorum Longus	Medial part of posterior tibia	Plantar surfaces of bases of distal phalanges of lateral four toes	Flexes lateral four toes; weakly plantarflexes foot	Tibial Nerve (S2, S3)
Tibialis Posterior	Interosseous membrane, posterior tibia and fibula	Navicular tuberosity, cuneiforms, cuboid, bases of metatarsals 2-4	Plantarflexes and inverts foot; supports medial arch	Tibial Nerve (L4, L5)

18.3 The Intrinsic Muscles of the Foot

The intrinsic muscles of the foot originate and insert within the foot itself. They are responsible for the fine motor movements of the toes and provide crucial dynamic support to the arches during gait. They are organized into a dorsal group and four plantar layers. **Dorsal Group** Located on the dorsum of the foot, this group is innervated by the **deep fibular nerve**. **Extensor Digitorum Brevis:** Extends toes 2-4. **Extensor Hallucis Brevis:** Extends the great toe. **Plantar Group** The ten intrinsic muscles on the plantar aspect of the foot are arranged in four layers, from superficial to deep. They are innervated by the **medial and lateral plantar nerves**, which are terminal branches of the tibial nerve. **First Layer (most superficial):** Abductor Hallucis, Flexor Digitorum Brevis, Abductor Digiti Minimi. **Second Layer:** Quadratus Plantae (which straightens the pull of the FDL tendon) and the four Lumbricals. **Third Layer:** Flexor Hallucis Brevis, Adductor Hallucis (with oblique and transverse heads), Flexor Digiti Minimi Brevis. **Fourth Layer (deepest):** Three Plantar Interossei (Adduct the toes - "PAD") and four Dorsal Interossei (Abduct the toes - "DAB").

Muscle	Origin	Insertion	Primary Action(s)	Innervation
Dorsal Group				
Extensor Digitorum Brevis	Superolateral surface of calcaneus	Extensor expansions of toes 2-4	Extends toes 2-4	Deep Fibular Nerve
Extensor Hallucis Brevis	Superolateral surface of calcaneus	Dorsal aspect of base of proximal phalanx of great toe	Extends great toe	Deep Fibular Nerve
Plantar Group - 1st Layer				
Abductor Hallucis	Medial tubercle of calcaneus, plantar aponeurosis	Medial side of base of proximal phalanx of great toe	Abducts and flexes great toe	Medial Plantar Nerve
Flexor Digitorum Brevis	Medial tubercle of calcaneus, plantar aponeurosis	Middle phalanges of lateral four toes	Flexes lateral four toes	Medial Plantar Nerve
Abductor Digiti Minimi	Medial and lateral tubercles of calcaneus	Lateral side of base of proximal phalanx of 5th toe	Abducts and flexes 5th toe	Lateral Plantar Nerve
Plantar Group - 2nd Layer				
Quadratus Plantae	Medial and lateral surfaces of calcaneus	Tendons of flexor digitorum longus	Assists FDL in flexing toes	Lateral Plantar Nerve

Muscle	Origin	Insertion	Primary Action(s)	Innervation
Lumbricals (4)	Tendons of flexor digitorum longus	Extensor expansions of lateral four toes	Flex MTP joints, extend IP joints	Medial 1: Medial Plantar N.; Lateral 3: Lateral Plantar N.
Plantar Group - 3rd Layer				
Flexor Hallucis Brevis	Cuboid and lateral cuneiforms	Medial and lateral sides of base of proximal phalanx of great toe	Flexes great toe's MTP joint	Medial Plantar Nerve
Adductor Hallucis	Oblique Head: Bases of metatarsals 2-4; Transverse Head: Plantar ligaments of MTP joints 3-5	Lateral side of base of proximal phalanx of great toe	Adducts great toe; supports arches	Deep branch of Lateral Plantar Nerve
Flexor Digiti Minimi Brevis	Base of 5th metatarsal	Base of proximal phalanx of 5th toe	Flexes 5th toe's MTP joint	Superficial branch of Lateral Plantar Nerve
Plantar Group - 4th Layer				
Plantar Interossei (3)	Medial sides of metatarsals 3-5	Medial sides of bases of proximal phalanges of toes 3-5	Adducts toes 3-5 (PAD)	Lateral Plantar Nerve
Dorsal Interossei (4)	Adjacent sides of metatarsals 1-5	Bases of proximal phalanges of toes 2-4 (1st medial, 2-4 lateral)	Abducts toes 2-4 (DAB)	Lateral Plantar Nerve

19 Key Anatomical Spaces and Clinical Implications

The lower limb contains several well-defined anatomical spaces through which major neurovascular structures pass. Understanding the boundaries and contents of these spaces is crucial for clinical practice.

19.1 The Femoral Triangle, Adductor Canal, and Popliteal Fossa

Femoral Triangle: An important anatomical region in the superoanterior thigh. Its boundaries are the inguinal ligament (superiorly), the sartorius muscle (laterally), and the adductor longus muscle (medially). From lateral to medial, its contents include the **femoral nerve, femoral artery, femoral vein**, and femoral lymphatics (a useful mnemonic is NAVEL: Nerve, Artery, Vein, Empty space, Lymphatics). The femoral pulse can be palpated here, and the femoral vein is a common site for central venous access. **Adductor Canal (Hunter's or Subsartorial Canal):** A narrow passageway in the middle third of the thigh, deep to the sartorius muscle. It provides a conduit for the femoral artery, femoral vein, and the saphenous nerve to travel from the femoral triangle to the popliteal fossa. It is bounded by the vastus medialis (anterolaterally), the adductor longus and magnus (posteriorly), and is roofed by the vastoadductor membrane and sartorius muscle. **Popliteal Fossa:** A diamond-shaped space on the posterior aspect of the knee. It is bounded by the biceps femoris (superolaterally), the semimembranosus and semitendinosus (superomedially), and the two heads of the gastrocnemius (inferiorly). It contains the **popliteal artery** (deepest), **popliteal vein**, and the **tibial and common fibular nerves** (most superficial).

19.2 Compartment Syndrome: Pathophysiology and Anatomical Basis

The leg is divided into four non-distensible osteofascial compartments: anterior, lateral, superficial posterior, and deep posterior. **Acute compartment syndrome** is a surgical emergency that occurs when pressure within one of these compartments rises to a level that compromises tissue perfusion. This is most commonly caused by trauma, such as a tibial fracture, leading to bleeding and edema within the confined space. As intracompartmental pressure exceeds capillary perfusion pressure, a vicious cycle of ischemia, cellular swelling, and further pressure increase ensues, leading to irreversible muscle and nerve damage (myoneural necrosis) if not treated promptly with a surgical **fasciotomy**. The anterior compartment is the most frequently affected site. Clinical signs are often remembered by the "6 Ps": Pain (out of proportion, especially with passive stretch), Paresthesia, Paresis, Pallor, Pulselessness, and Poikilothermia (coolness). Paresthesia and severe pain on passive stretch are the earliest and most reliable signs.

19.3 Trendelenburg's Sign: The Anatomy of Pelvic Stabilization

Trendelenburg's sign is a key physical examination finding used to assess the function of the hip abductor muscles, specifically the gluteus medius and minimus, and the integrity of their innervation by the superior gluteal nerve. During a normal gait cycle, when one leg is lifted off the ground (swing phase), the gluteus medius and minimus on the opposite (stance) leg contract powerfully. This contraction prevents the pelvis from tilting down on the unsupported side. If these muscles are weak or paralyzed (e.g., due to a superior gluteal nerve injury), they cannot stabilize the pelvis. When the patient is asked to stand on the affected leg, the pelvis will drop on the contralateral (unsupported) side. This is a positive Trendelenburg sign and results in a characteristic "waddling" or "gluteal" gait. The convergence

of three muscles from three different compartments—the sartorius (anterior), gracilis (medial), and semitendinosus (posterior)—onto a single insertion point on the anteromedial tibia known as the **pes anserinus** (“goose’s foot”) is a remarkable anatomical feature. This arrangement means that three distinct major nerves (femoral, obturator, and tibial) control muscles that act on this single point, providing multiplanar dynamic stability to the medial aspect of the knee. Clinically, the bursa located deep to these tendons, the **pes anserine bursa**, can become inflamed, leading to medial knee pain. The differential diagnosis for this condition must consider pathology originating from any of the three thigh compartments that contribute to this tendinous insertion.

Part VII

The Neurovascular Network

The lower limb is supplied by a complex and extensive network of nerves, arteries, and veins that originate in the lumbar and pelvic regions and extend to the tips of the toes. These structures are often bundled together and are vulnerable to injury at specific anatomical “choke points.”

20 Innervation of the Lower Limb

The entire nerve supply to the lower limb originates from the lumbosacral plexus, a network of nerves formed by the anterior rami of the lumbar and sacral spinal nerves.

20.1 The Lumbosacral Plexus: Formation and Major Nerves

The lumbosacral plexus is formed from the anterior rami of spinal nerves L1 through S4, with a contribution from T12. It is not a single entity but is typically described as two interconnected plexuses: the lumbar plexus and the sacral plexus. **Lumbar Plexus (L1-L4):** This plexus is formed within the substance of the psoas major muscle on the posterior abdominal wall. Its major branches emerge to supply the anterior and medial thigh, as well as the skin of the anteromedial leg. **Sacral Plexus (L4-S4):** This plexus is formed in the pelvis, anterior to the piriformis muscle. It is connected to the lumbar plexus via the lumbosacral trunk (formed by the anterior rami of L4 and L5). Its branches supply the gluteal region, the posterior thigh, and the entire leg and foot.

Nerve	Spinal Roots	Motor Innervation (Key Muscle Groups)	Sensory Innervation (Cutaneous Territory)
Lumbar Plexus			
Femoral Nerve	L2, L3, L4	Anterior thigh muscles (quadriceps, sartorius, iliacus, pectineus)	Anterior thigh (anterior cutaneous branches); medial leg and foot (saphenous nerve)
Obturator Nerve	L2, L3, L4	Medial thigh muscles (adductors, gracilis)	Small area on medial thigh
Lateral Femoral Cutaneous Nerve	L2, L3	None	Anterolateral surface of the thigh
Sacral Plexus			
Superior Gluteal Nerve	L4, L5, S1	Gluteus medius, gluteus minimus, tensor fasciae latae	None
Inferior Gluteal Nerve	L5, S1, S2	Gluteus maximus	None
Sciatic Nerve (Tibial & Common Fibular)	L4, L5, S1, S2, S3	Posterior thigh muscles (hamstrings); all muscles of the leg and foot	Posterior leg; lateral leg; entire foot (except small medial area by saphenous n.)
Posterior Femoral Cutaneous Nerve	S1, S2, S3	None	Posterior surface of thigh and upper leg
Pudendal Nerve	S2, S3, S4	Muscles of the perineum	Skin of the perineum and external genitalia

20.2 Course and Distribution of Major Nerves

Femoral Nerve (L2-L4): After forming in the psoas major, it descends to enter the thigh by passing deep to the inguinal ligament, within the femoral triangle, lateral to the femoral artery. It quickly divides into numerous muscular branches for the anterior thigh and cutaneous branches for the anterior thigh skin. Its largest and longest branch, the **saphenous nerve**, continues through the adductor canal to provide sensory innervation to the medial aspect of the leg and foot. **Obturator Nerve (L2-L4):** This nerve emerges from the medial border of the psoas major, travels through the pelvis, and enters the medial thigh by passing through the obturator canal. Here, it divides into anterior and posterior branches that pass around the adductor brevis muscle to supply the adductor group. **Sciatic Nerve (L4-S3):** The largest nerve in the body, it exits the pelvis through the greater sciatic foramen, typically inferior to the piriformis muscle. It descends deep in the posterior thigh, supplying the hamstring muscles. At the apex of the popliteal fossa, it divides into its two terminal branches: **Tibial Nerve:** The larger medial division, it continues inferiorly through the popliteal fossa and the deep posterior compartment of the leg, innervating all posterior leg muscles. It then passes through the tarsal tunnel posterior to the medial malleolus and divides into the **medial and lateral plantar nerves**, which supply all intrinsic muscles and the skin of the sole of the foot.

Common Fibular (Peroneal) Nerve: The smaller lateral division, it follows the medial border of the biceps femoris tendon and winds superficially around the neck of the fibula. This exposed position makes it the most commonly injured nerve in the lower limb. It then divides into the **superficial fibular nerve** (for the lateral compartment) and the **deep fibular nerve** (for the anterior compartment). The neurovascular bundles of the lower limb are particularly susceptible to compression or injury at specific "choke points" where they traverse unyielding osteofibrous tunnels or lie in close proximity to mobile structures. The sciatic nerve's passage deep to the piriformis muscle is the basis for piriformis syndrome. The common fibular nerve's superficial course around the fibular neck makes it vulnerable to direct trauma, fracture, or pressure from a tight cast, leading to foot drop. The tibial nerve's passage through the tarsal tunnel is the site of entrapment in tarsal tunnel syndrome. A thorough understanding of these anatomical vulnerabilities is fundamental to diagnosing compression neuropathies.

20.3 Cutaneous Innervation and Dermatome Mapping

The skin of the lower limb is supplied by cutaneous branches of the major peripheral nerves. It is important to distinguish the area of skin supplied by a single peripheral nerve (its cutaneous field) from a **dermatome** (the area of skin supplied by a single spinal cord segment). **Anterior Thigh:** Supplied by anterior cutaneous branches of the femoral nerve. **Lateral Thigh:** Supplied by the lateral femoral cutaneous nerve. **Medial Thigh:** Supplied by the obturator nerve and medial cutaneous branches of the femoral nerve. **Posterior Thigh:** Supplied by the posterior femoral cutaneous nerve. **Medial Leg and Foot:** Supplied by the saphenous nerve (from the femoral nerve). **Lateral Leg and Dorsum of Foot:** Supplied by the superficial fibular nerve. **Webspace between 1st and 2nd Toes:** Supplied by the deep fibular nerve. **Sole of the Foot:** Supplied by the medial and lateral plantar nerves (from the tibial nerve). The shared root values of nerves supplying different regions explain the phenomenon of referred pain. For example, the femoral nerve (L2-L4) and obturator nerve (L2-L4) both send articular branches to the hip and knee joints. Consequently, pathology in the hip joint, such as osteoarthritis, can irritate these nerves and manifest as pain perceived in the knee. Similarly, compression of the L4 nerve root in the lumbar spine can cause pain to radiate along the dermatome down the anterior thigh and medial leg, mimicking local pathology. This anatomical principle underscores the importance of examining the joint above and below the site of pain, as well as the corresponding spinal levels.

21 Vasculature of the Lower Limb

21.1 Arterial Supply: From the Femoral Artery to the Plantar Arch

The arterial supply to the lower limb originates from the **external iliac artery**, which becomes the **femoral artery** as it passes deep to the inguinal ligament to enter the femoral triangle. **In the Thigh:** The femoral artery is the main arterial trunk. Its largest branch, the **profunda femoris (deep femoral) artery**, arises in the femoral triangle and is the chief artery to the thigh. It gives off **medial and lateral circumflex femoral arteries**, which form an anastomotic ring around the femoral neck and are the primary blood supply to the femoral head, and several perforating arteries that supply the posterior compartment. The femoral artery itself continues down the thigh through the adductor canal. **At the Knee:** After passing through the adductor hiatus (an opening in the adductor magnus), the femoral artery becomes the **popliteal artery**. In the popliteal fossa, it gives off several **genicular arteries** that form a rich anastomosis around the knee joint. **In the Leg:** At the inferior border of the popliteus muscle, the popliteal artery divides into the **anterior tibial artery** and the **posterior tibial artery**. The anterior tibial artery passes through an opening in the interosseous membrane to supply the anterior compartment. The posterior tibial artery descends in the posterior compartment, giving off the **fibular (peroneal) artery**, which supplies the lateral compartment. **In the Foot:** The anterior tibial artery continues onto the dorsum of the foot as the **dorsalis pedis artery**. The posterior tibial artery passes behind the medial malleolus, through the tarsal tunnel, and divides into the **medial and lateral plantar arteries** in the sole of the foot. The lateral plantar artery and the dorsalis pedis artery anastomose to form the **deep plantar arch**, which supplies the digits.

21.2 Venous Drainage: The Superficial and Deep Systems

The venous drainage of the lower limb is organized into two interconnected systems: superficial and deep. **Superficial Veins:** These veins are located within the subcutaneous tissue (superficial fascia). The two principal superficial veins are: **Great Saphenous Vein:** The longest vein in the body, it begins on the medial side of the dorsal venous arch of the foot, ascends anterior to the medial malleolus, and runs up the medial side of the leg and thigh. It drains into the femoral vein at the saphenous opening in the femoral triangle. It is commonly harvested for coronary artery bypass grafting. **Small Saphenous Vein:** Begins on the lateral side of the dorsal venous arch, ascends posterior to the lateral malleolus, and runs up the posterior aspect of the leg. It typically drains into the popliteal vein in the popliteal fossa. **Deep Veins:** These veins accompany the major arteries and are located deep to the muscular fascia. They are named according to the artery they travel with (e.g., anterior and posterior tibial veins, popliteal vein, femoral vein). **Perforating Veins:** These vessels connect the superficial and deep venous systems. They contain one-way valves that direct blood flow from the superficial to the deep system. The contraction of the powerful calf muscles during walking (the **musculovenous pump**) squeezes the deep veins,

propelling blood towards the heart against gravity.

21.3 Lymphatics

The lymphatic drainage of the lower limb generally follows the venous system. Superficial lymphatic vessels, particularly those accompanying the great saphenous vein, drain into the **superficial inguinal lymph nodes**. Deep lymphatic vessels, which follow the deep veins, drain into the **deep inguinal lymph nodes**. From the inguinal nodes, lymph passes to the external iliac nodes and then into the lumbar lymphatic trunks.

Part VIII

Integrated Clinical Anatomy and Procedures

This final part synthesizes the foundational anatomical knowledge to provide a deeper understanding of common clinical conditions, physical examination techniques, and procedures related to the lower limb.

22 Common Pathologies: An Anatomical Perspective

22.1 Fractures and Dislocations

Femoral Neck Fractures Fractures of the femoral neck are common, particularly in elderly individuals with osteoporosis, and are classified as intracapsular hip fractures. Their clinical significance is profoundly influenced by the vascular anatomy of the femoral head. The primary blood supply comes from the **medial femoral circumflex artery**, which gives rise to **retinacular arteries** that ascend along the femoral neck to supply the head. A displaced fracture of the femoral neck can tear these delicate vessels, disrupting the blood supply. This places the femoral head at high risk for **avascular necrosis (AVN)**, a condition where the bone tissue dies, leading to collapse of the head and severe arthritis. This vascular vulnerability is the primary reason why displaced femoral neck fractures in the elderly are often treated with hip replacement (arthroplasty) rather than internal fixation. **Tibial Shaft Fractures** The tibia is the most frequently fractured long bone in the body. The anteromedial surface of the tibia is subcutaneous along its entire length, making it highly susceptible to **open (compound) fractures**, where the bone penetrates the skin. Open fractures carry a high risk of infection (osteomyelitis) and wound healing complications. The high-energy forces often required to fracture the tibia can lead to

extensive soft tissue damage and comminuted fracture patterns, and they are the most common cause of acute compartment syndrome. **Hip Dislocation** Traumatic hip dislocations are severe injuries that require immense force, such as that from a motor vehicle collision. In approximately 90% of cases, the dislocation is **posterior**, where the femoral head is forced backward out of the acetabulum. This typically occurs from a force applied to a flexed, adducted hip, such as the knee striking a dashboard. Because the **sciatic nerve** passes directly posterior to the hip joint, it is at high risk of being stretched or compressed during a posterior dislocation, which can result in foot drop and sensory loss. Furthermore, the dislocation can tear the retinacular arteries that supply the femoral head, leading to a significant risk of AVN, similar to that seen with femoral neck fractures. **Patellar Dislocation** Patellar dislocations almost always occur in a **lateral** direction. The primary anatomical restraint against lateral displacement is the **medial patellofemoral ligament (MPFL)**, which tethers the medial border of the patella to the medial femoral condyle. A first-time traumatic dislocation typically involves a rupture of the MPFL. Several anatomical factors can predispose an individual to patellar instability, including a shallow trochlear groove on the femur (trochlear dysplasia), a high-riding patella (patella alta), and malalignment of the lower limb that increases the **Q-angle** (the angle between the line of pull of the quadriceps and the patellar tendon).

22.2 Soft Tissue Injuries

ACL and Meniscal Tears The anterior cruciate ligament (ACL) is most commonly torn during non-contact sports that involve sudden deceleration, pivoting, or changes in direction. The injury often occurs with a valgus and rotational force applied to the knee, which can also damage the MCL and medial meniscus, resulting in the "unhappy triad". Meniscal tears can be traumatic (in younger individuals, often from twisting on a flexed knee) or degenerative (in older individuals). The healing potential of a meniscal tear is dictated by its location relative to its blood supply. The outer one-third of the meniscus is vascularized (the "red zone"), while the inner two-thirds are avascular (the "white zone"). Tears in the red zone have healing potential, whereas tears in the white zone do not and may require surgical resection (meniscectomy). **Ankle Sprains** Ankle sprains are among the most common musculoskeletal injuries. The vast majority are **inversion sprains** that damage the lateral ligament complex. The **anterior talofibular ligament (ATFL)** is the most frequently injured ligament, typically torn during a combined inversion and plantarflexion movement. More severe inversion injuries can also involve the **calcaneofibular ligament (CFL)** and, rarely, the **posterior talofibular ligament (PTFL)**.

22.3 Systemic and Overuse Conditions

Deep Vein Thrombosis (DVT) DVT is the formation of a blood clot (thrombus) within the deep veins of the lower limb, most commonly in the calf veins (posterior tibial and peroneal veins). The condition is predisposed by **Virchow's triad**: venous stasis, endothelial injury, and a hypercoagulable state. Stasis is a major factor, often resulting from prolonged immobility (e.g., long flights, bed

rest after surgery), which inactivates the musculovenous pump of the calf. The primary danger of a DVT is that the clot can dislodge, travel through the venous system to the heart, and become lodged in the pulmonary arteries, causing a life-threatening **pulmonary embolism**. **Peripheral Artery Disease (PAD)** PAD is an atherosclerotic condition that causes narrowing (stenosis) of the arteries supplying the lower limbs. This narrowing reduces blood flow to the muscles and tissues. The most common symptom is **intermittent claudication**, which is cramping pain in the muscles (typically the calf) that is induced by exercise and relieved by rest. The location of the pain corresponds to the muscles distal to the arterial blockage; for example, stenosis in the femoral artery in the adductor canal often causes claudication in the calf muscles (gastrocnemius and soleus). Severe PAD can lead to pain at rest, non-healing ulcers, and gangrene. **Osteoarthritis** Osteoarthritis is a degenerative joint disease characterized by the breakdown of articular cartilage and changes in the underlying bone. In the lower limb, it most commonly affects the major weight-bearing joints: the hip and the knee. Risk factors include age, obesity, and previous joint injury. The condition leads to pain, stiffness, and reduced mobility, and in severe cases, may necessitate joint replacement surgery.

23 Applied Surface Anatomy and Clinical Procedures

23.1 Surface Anatomy and Palpation

A physical examination of the lower limb relies on the palpation of key bony landmarks to identify underlying structures. These include: **Hip/Pelvis:** Iliac crest, ASIS, PSIS, greater trochanter, pubic tubercle. **Knee:** Patella, tibial tuberosity, Gerdy's tubercle, head of the fibula, medial and lateral joint lines, femoral and tibial condyles. **Ankle/Foot:** Medial malleolus, lateral malleolus, navicular tuberosity, tuberosity of the fifth metatarsal, calcaneus.

23.2 Clinical Examination Tests

Anterior Drawer Test of the Knee: This test assesses the integrity of the ACL. With the patient supine and the knee flexed to 90 degrees, the examiner attempts to pull the tibia forward on the femur. Excessive anterior translation compared to the uninjured side indicates a likely ACL tear. The test directly challenges the primary function of the ACL, which is to prevent this anterior movement. **Trendelenburg's Sign:** As described previously, this test assesses the strength of the gluteus medius and minimus muscles. The patient stands on one leg. If the pelvis drops on the unsupported side, the sign is positive, indicating weakness of the abductors on the stance side, which points to a potential superior gluteal nerve lesion.

23.3 Pulse Points

Assessing arterial circulation in the lower limb is a fundamental clinical skill. The four main pulse points are palpated at specific anatomical locations: **Femoral Pulse:** Palpated in the femoral triangle, midway between the ASIS and the pubic symphysis (the mid-inguinal point). **Popliteal Pulse:** Palpated deep within the popliteal fossa, often with the patient's knee slightly flexed to relax the overlying fascia. It is the most difficult pulse to locate. **Posterior Tibial Pulse:** Palpated in the groove between the posterior aspect of the medial malleolus and the calcaneal tendon. **Dorsalis Pedis Pulse:** Palpated on the dorsum of the foot, just lateral to the tendon of the extensor hallucis longus.

23.4 Intramuscular Injections

The gluteal region is a common site for intramuscular injections. Safe administration requires precise anatomical knowledge to avoid injuring the sciatic nerve and superior gluteal vessels. **Dorsogluteal Site:** The traditional site is the upper outer quadrant of the buttock. This location is intended to be superior and lateral to the course of the sciatic nerve. **Ventrogluteal Site:** A safer alternative site. The landmarking involves placing the palm of the hand on the patient's greater trochanter, with the index finger on the ASIS and the middle finger spread posteriorly along the iliac crest. The injection is given in the "V" formed by the fingers, into the substance of the gluteus medius muscle, away from any major nerves or vessels. **Vastus Lateralis:** The anterolateral thigh is another common site, particularly in infants, using the vastus lateralis muscle.